

模拟电子技术基础

Analog Device

6 Feedback amplifier circuit

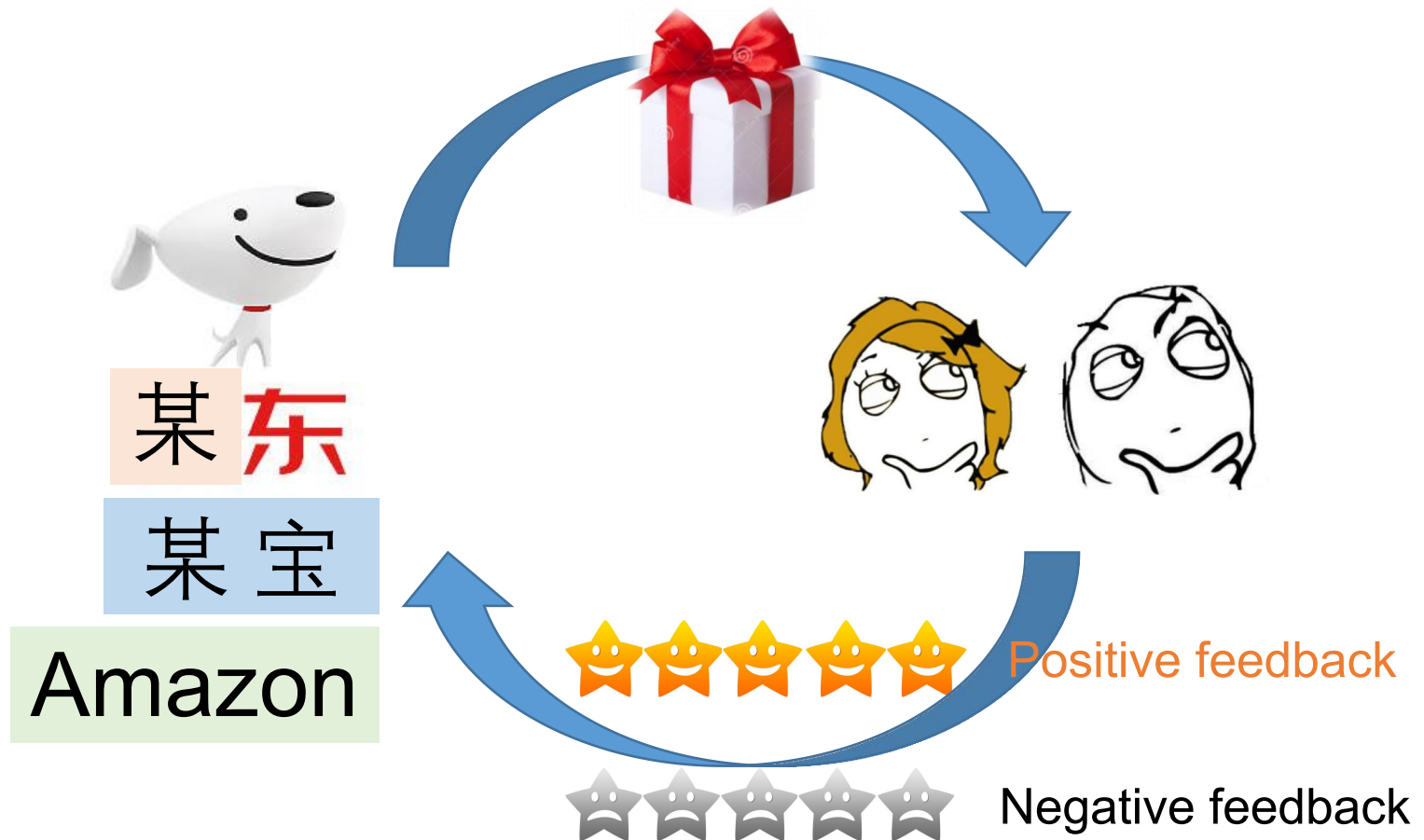
反馈放大电路

陈晓龙 Chen, Xiaolong

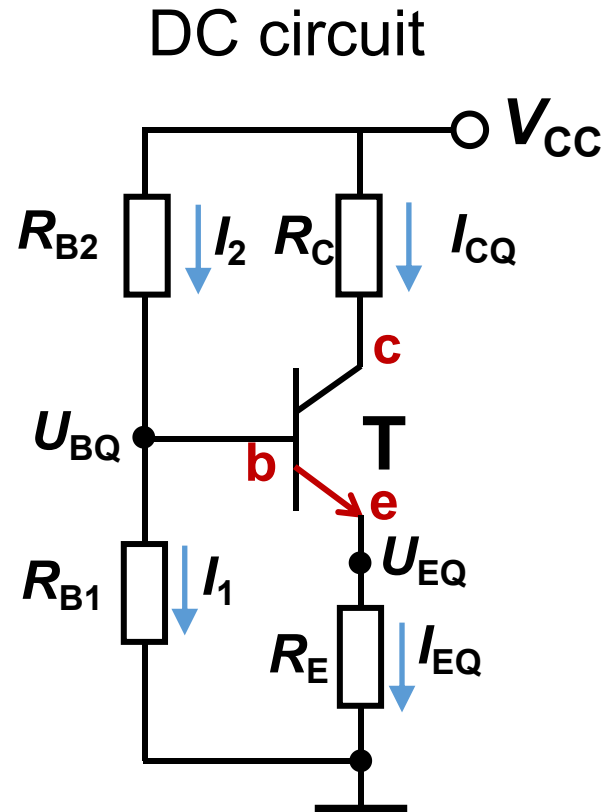
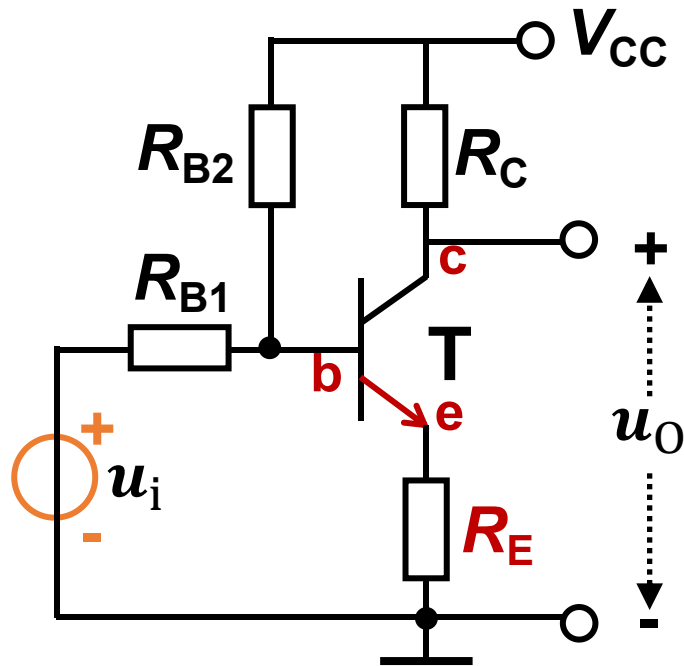
Department of Electrical and Electronic Engineering

6.0 What is feedback

An example of feedback in our daily life



Application of negative feedback in stabilizing the static working point Q



Working principle

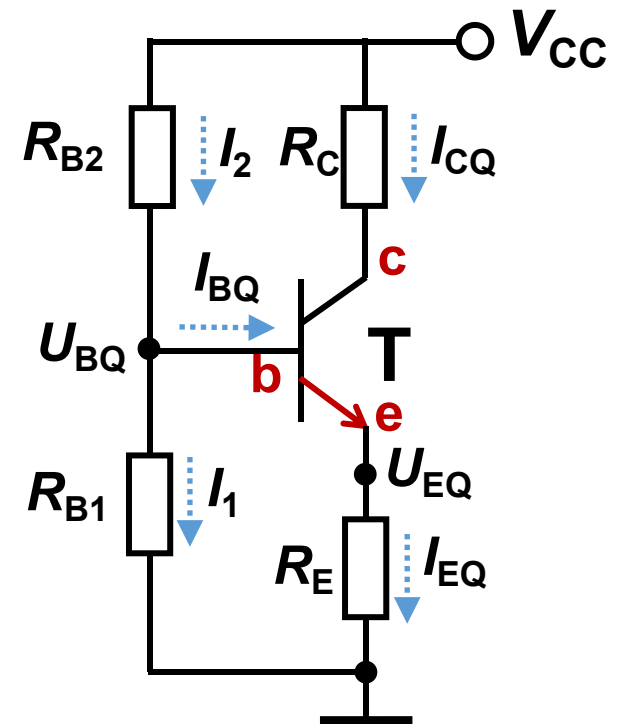
$$I_2 = I_1 + I_{BQ}$$

To make Q point stable, we let

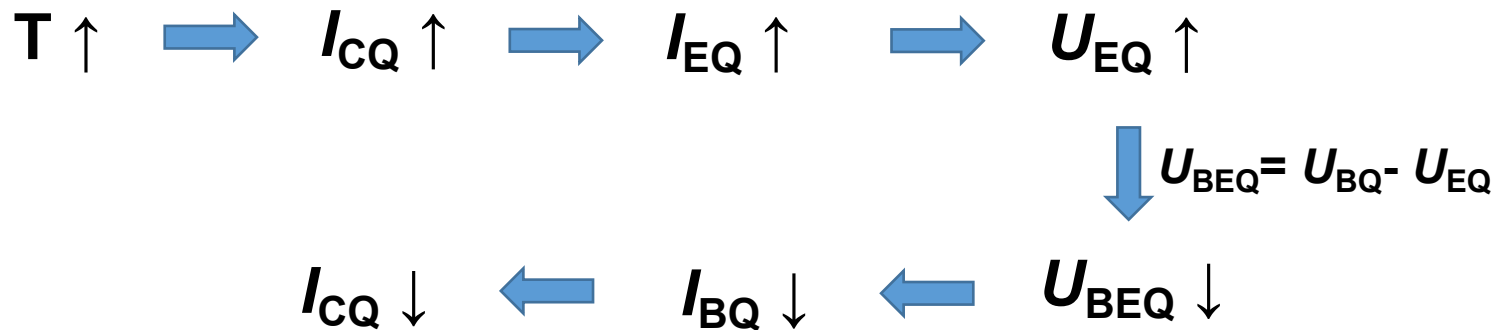
$$I_2 \text{ and } I_1 \gg I_{BQ}$$

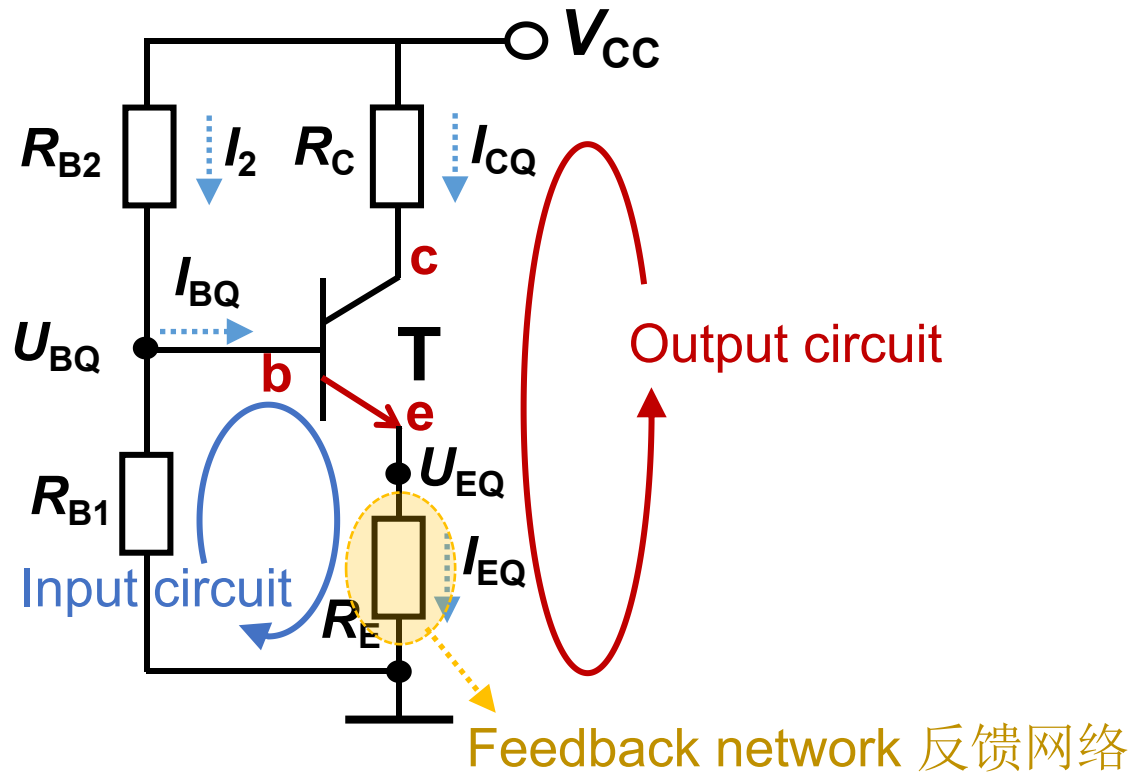
$$I_2 \approx I_1$$

$$U_{BQ} \approx \frac{R_{B1}}{R_{B1} + R_{B2}} \cdot V_{CC}$$



Then U_{BQ} is unrelated to temperature





Feedback 反馈:

In electronic circuits, the output signal is transported back to the input circuit, and influence the input signal.

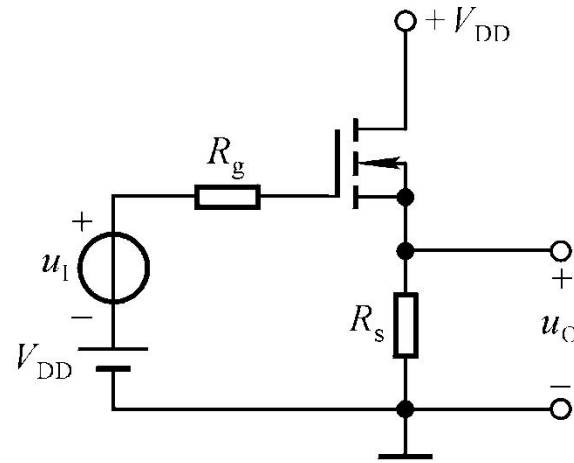
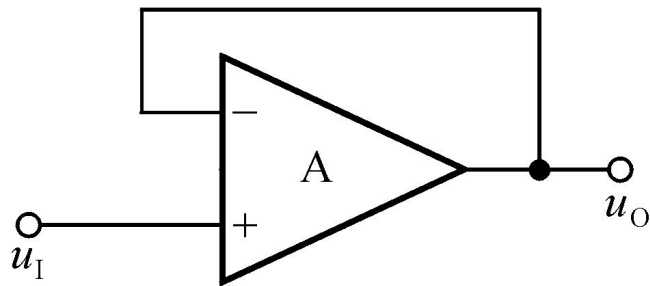
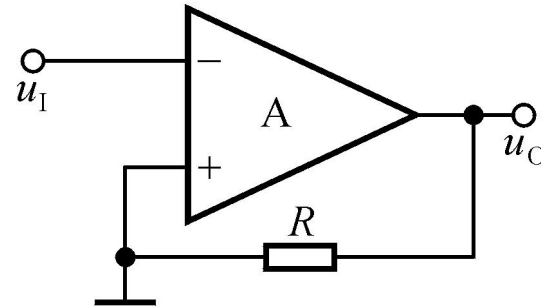
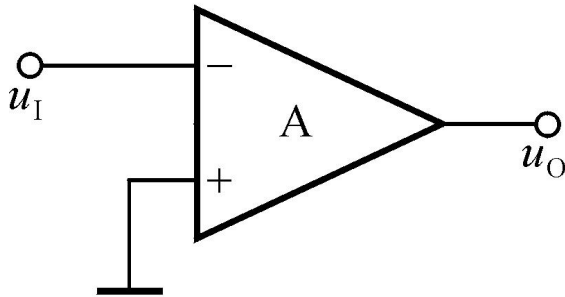
Feedback types

{ Positive feedback 正反馈
Negative feedback 负反馈

{ DC feedback 直流反馈
AC feedback 交流反馈

6.1 Judgment method of feedback types

Whether there is feedback or not? 有无反馈?



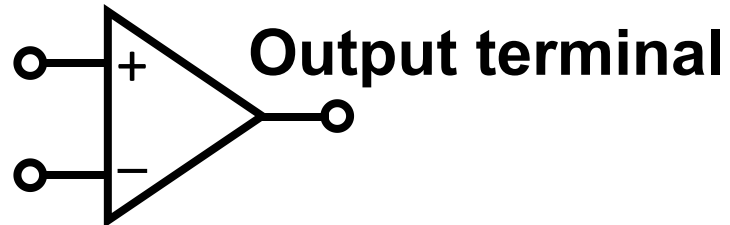
Electronic symbol of integrated operational amplifier 集成运放符号

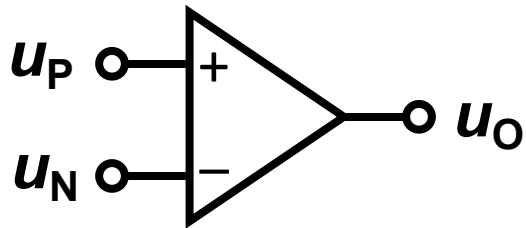
Non-inverting input terminal

同相输入端

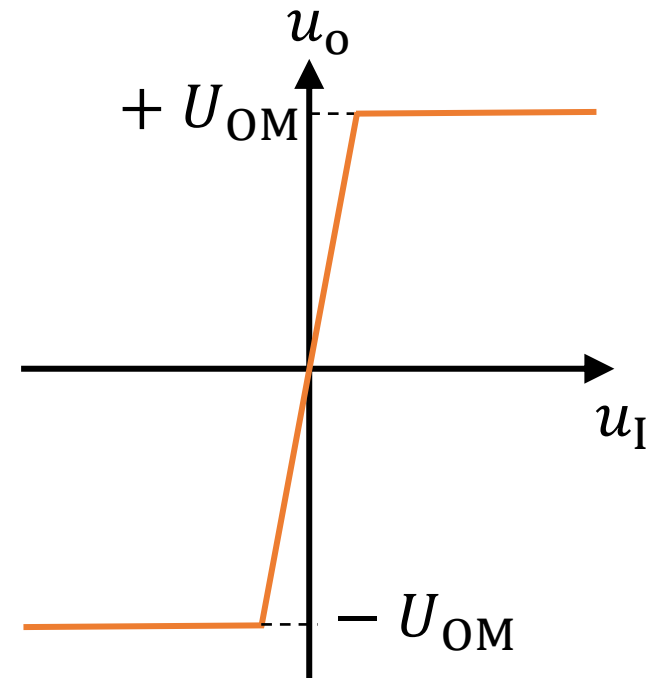
反相输入端

Inverting input terminal





Input signal $u_{Id} = u_P - u_N$

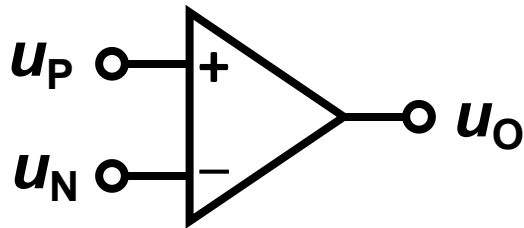


Linear amplification region

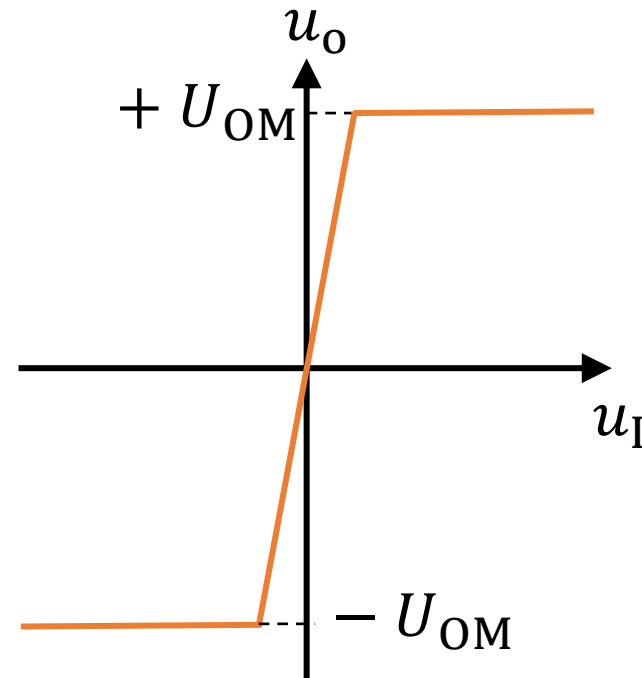
$$u_O = A_{od}(u_P - u_N)$$

A_{od} : Open loop differential mode voltage gain 开环差模电压增益

In IOC, A_{od} can be $>100,000$



$$u_{Id} = u_P - u_N$$



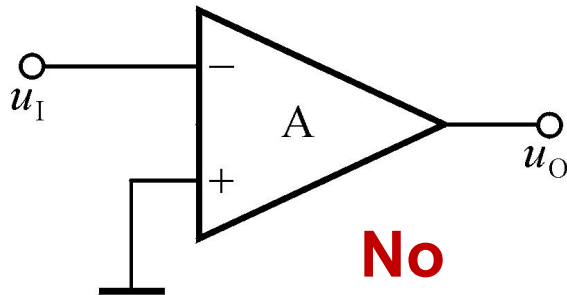
Saturation region

$$u_O = +U_{OM} \text{ or } -U_{OM}$$

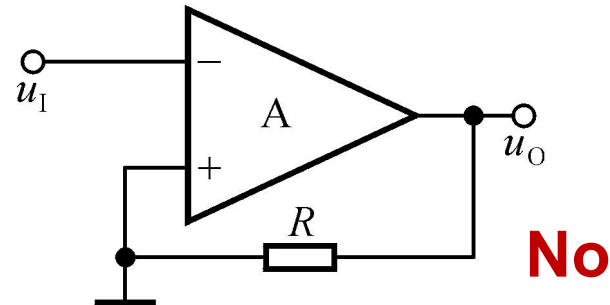
If $U_{OM}=14 \text{ V}$ and $A_{od}=500,000$, integrated operational amplifier enters saturation region when $|u_P - u_N| > 28 \mu\text{V}$

Whether there is feedback or not?

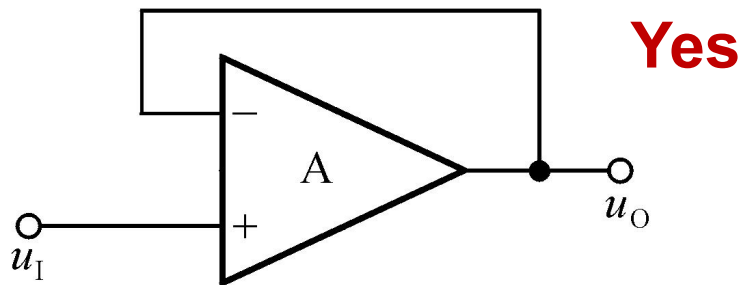
If there is FEEDBACK, output signal should influence the input signal



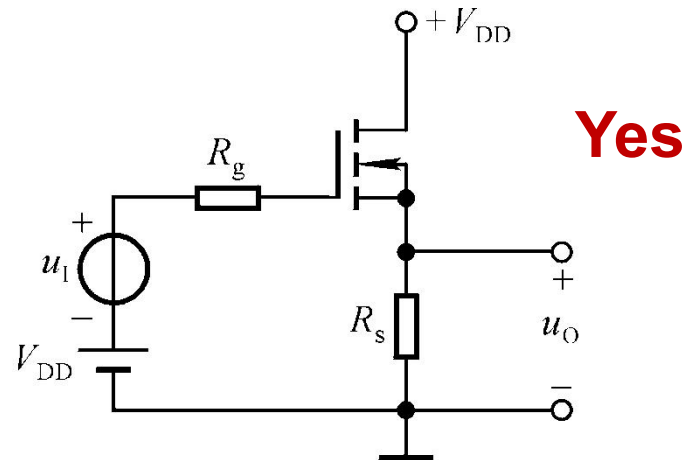
No



No



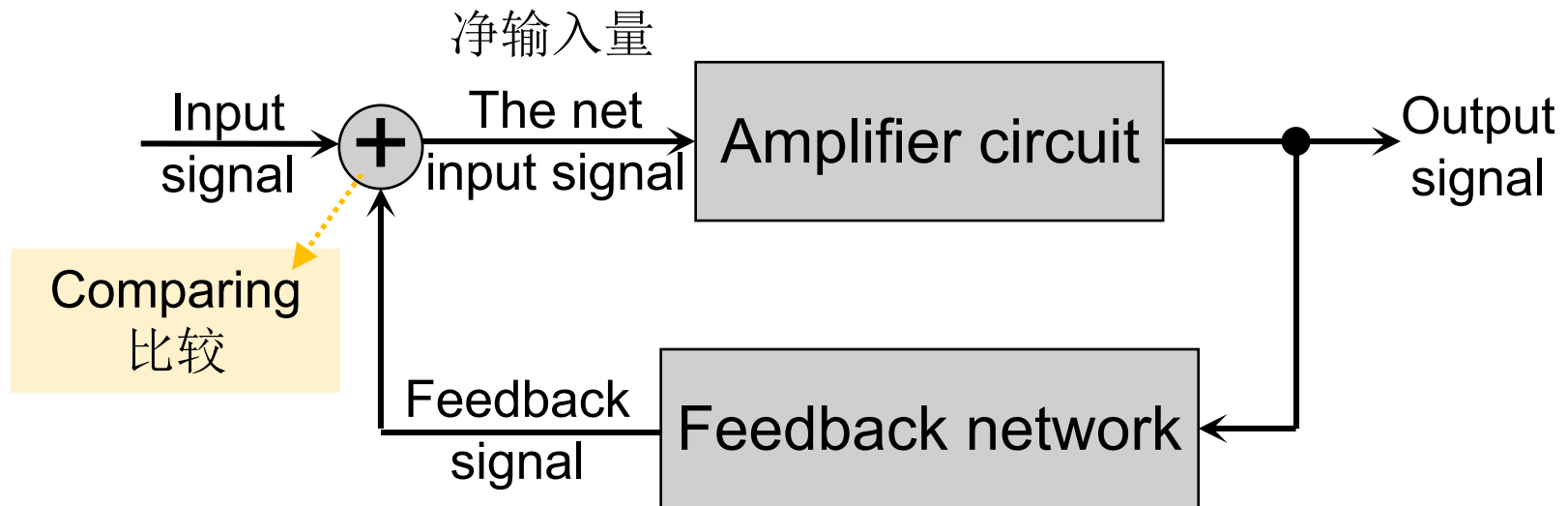
Yes



Yes

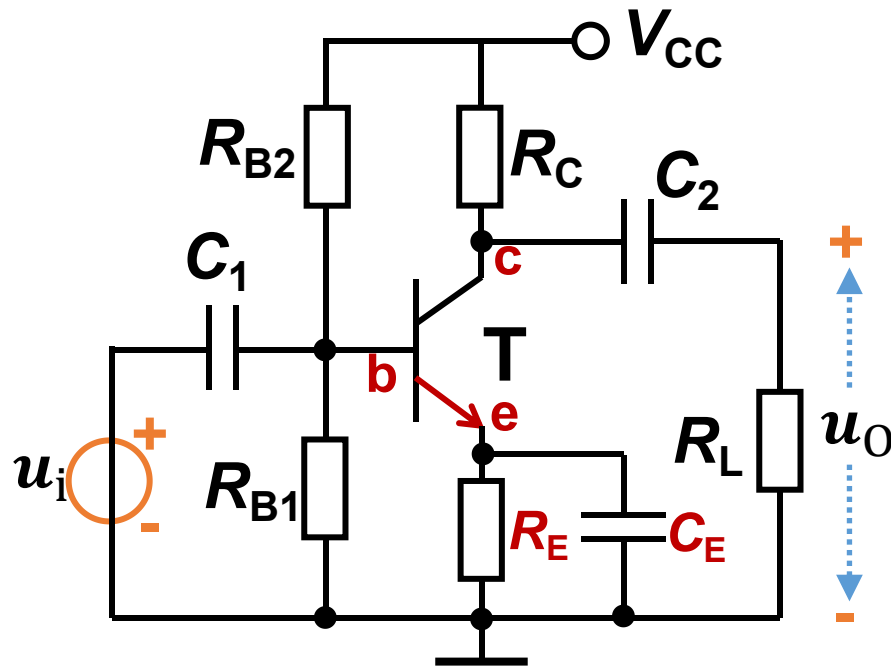
The block diagram of feedback amplifier circuit

反馈放大电路方框图

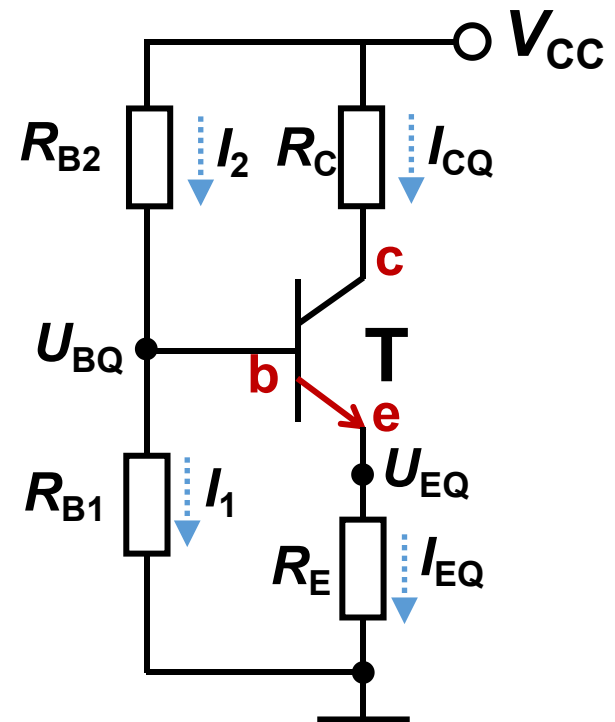


How to judge the DC and AC feedback 判断直流和交流反馈

Example 1:

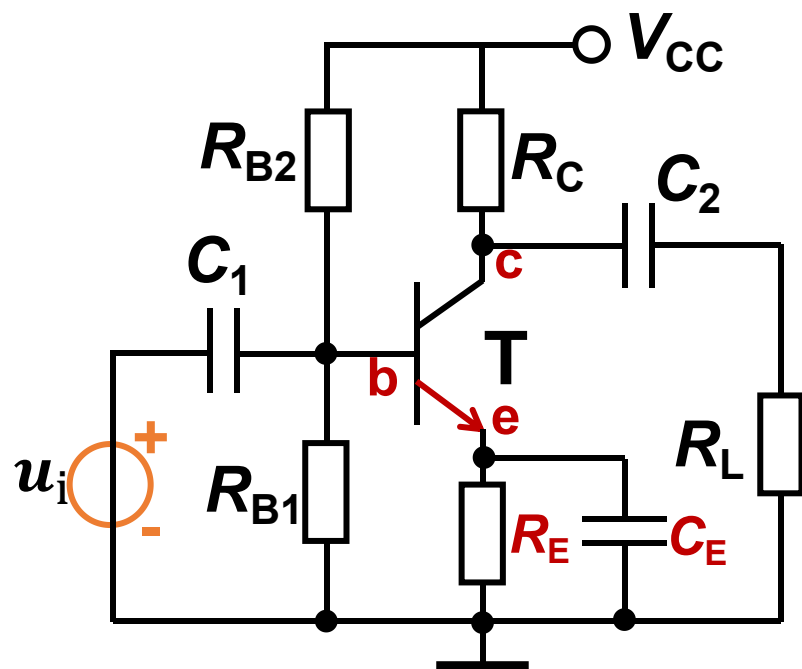


DC circuit

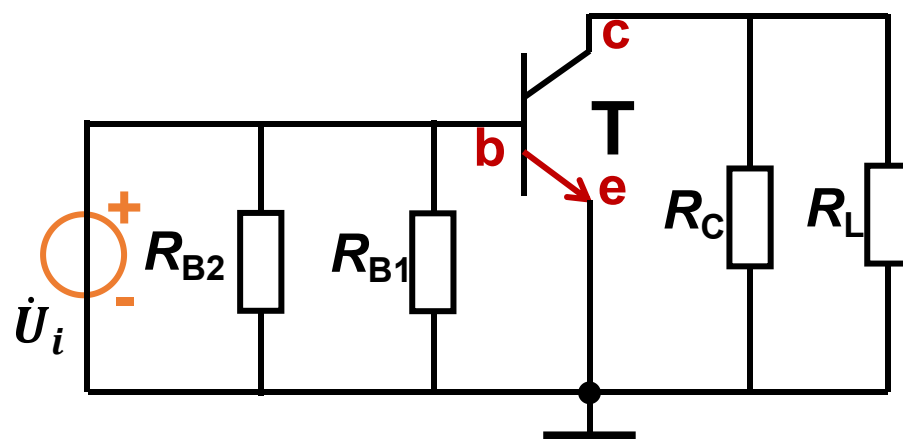


There is DC feedback!

Example 1:

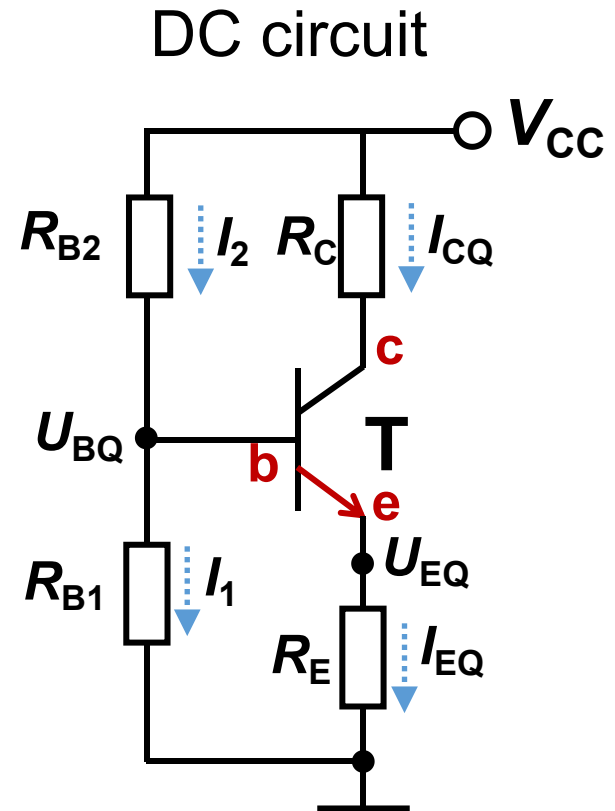
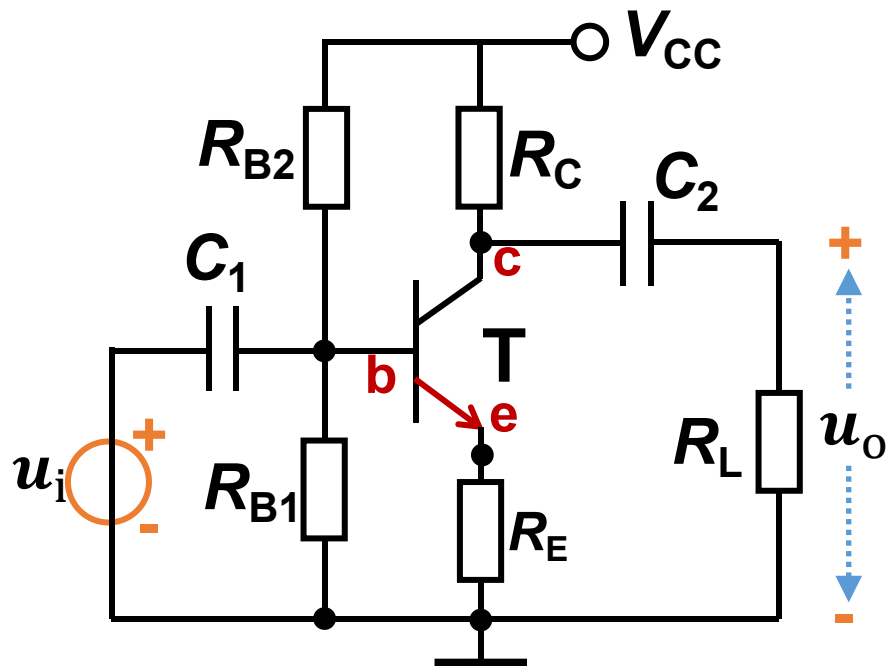


AC circuit



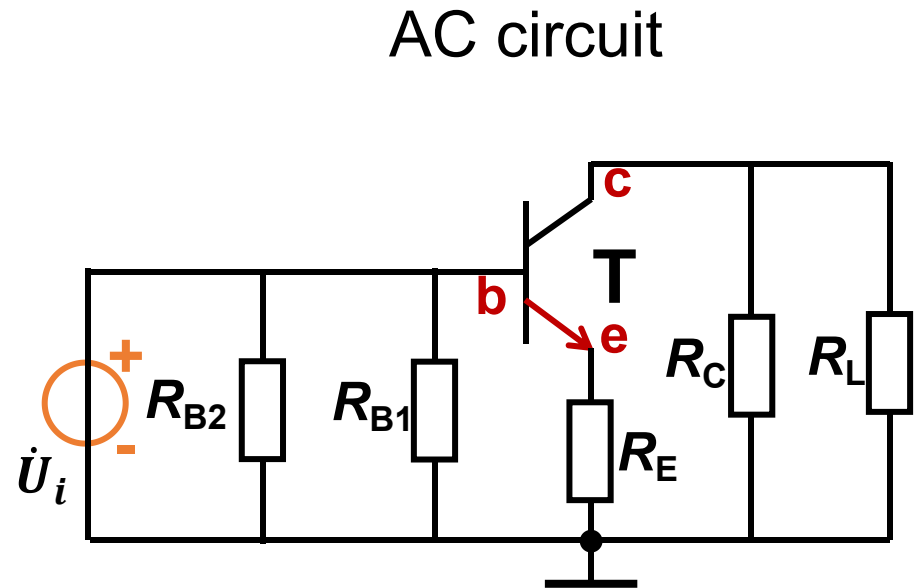
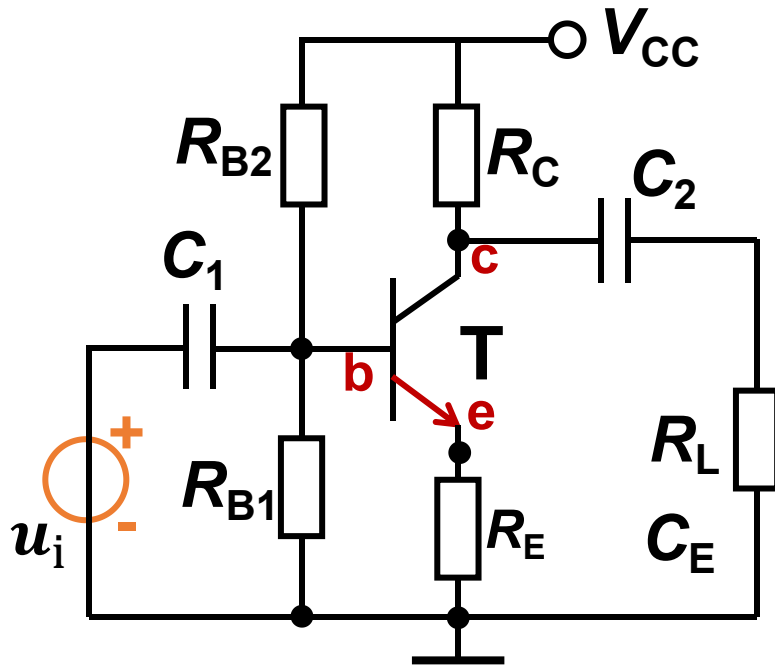
There is no AC feedback!

Example 2: Do DC and AC feedback exist in the circuit?



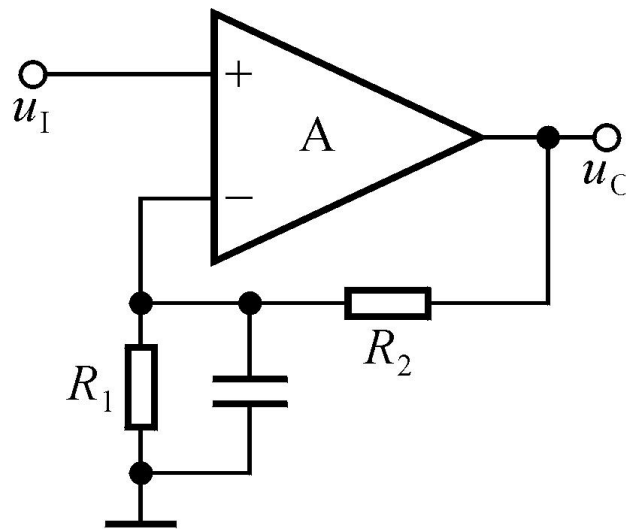
There is DC feedback!

Example 2: if there is AC feedback?

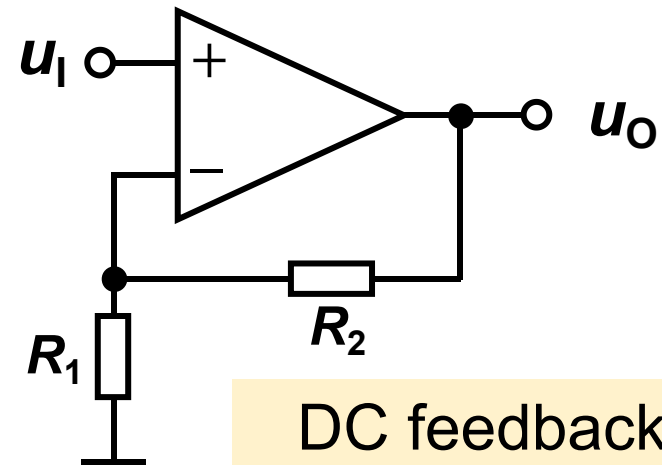


There is AC feedback!

Example 3:

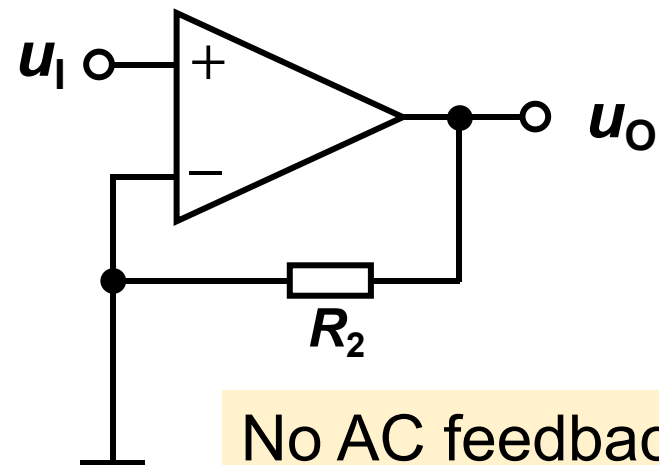


DC circuit



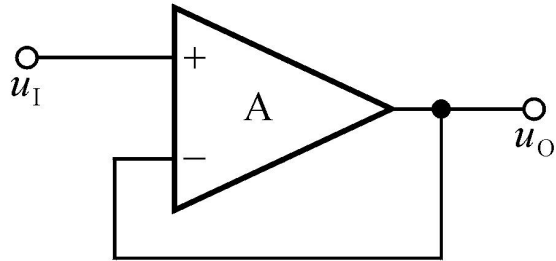
DC feedback!

AC circuit

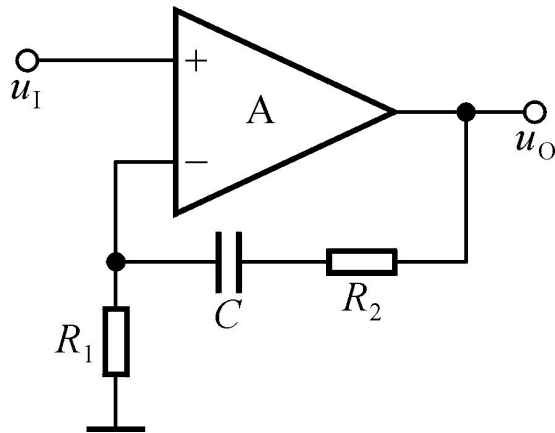


No AC feedback!

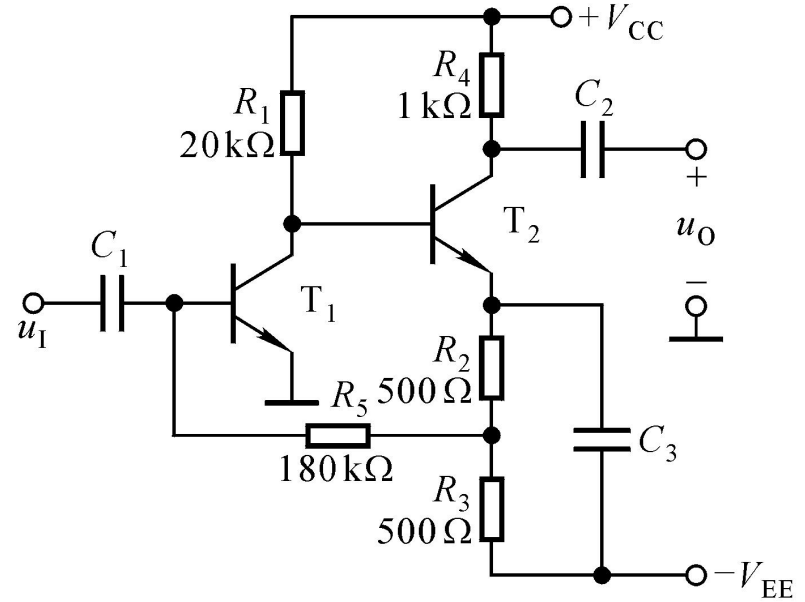
Example 4:



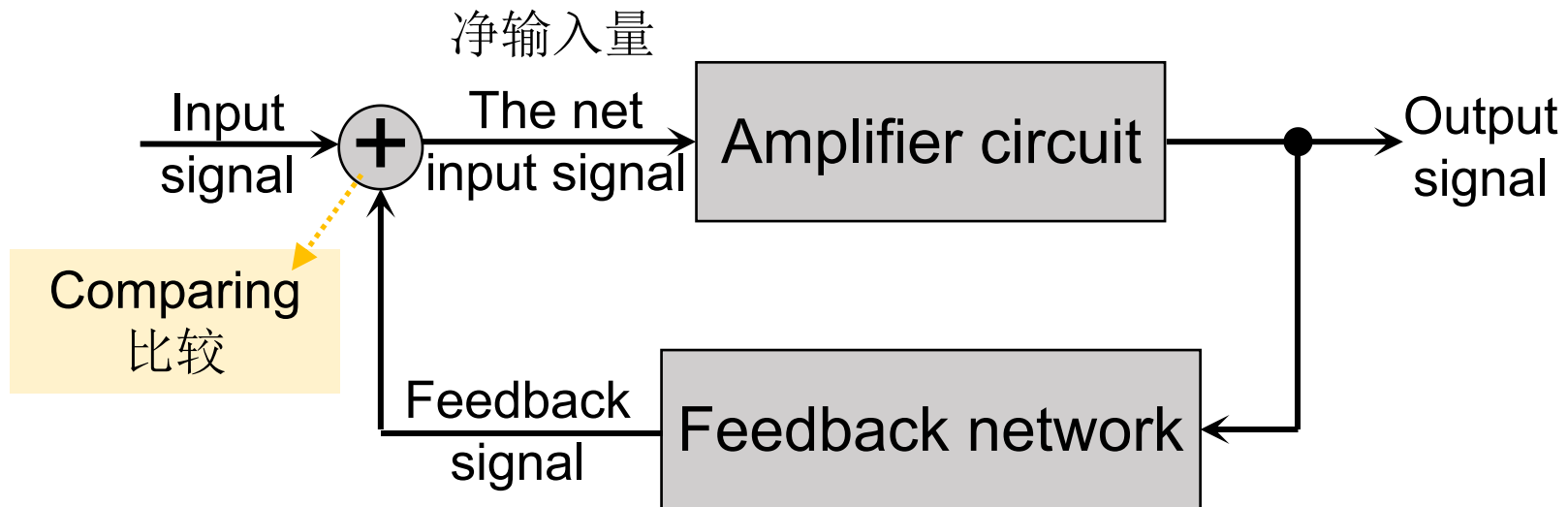
Example 5:



Example 6:



Positive and negative feedback 正反馈与负反馈



- The net input signal $>$ input signal: **positive feedback**

Can be applied in oscillator 振荡器

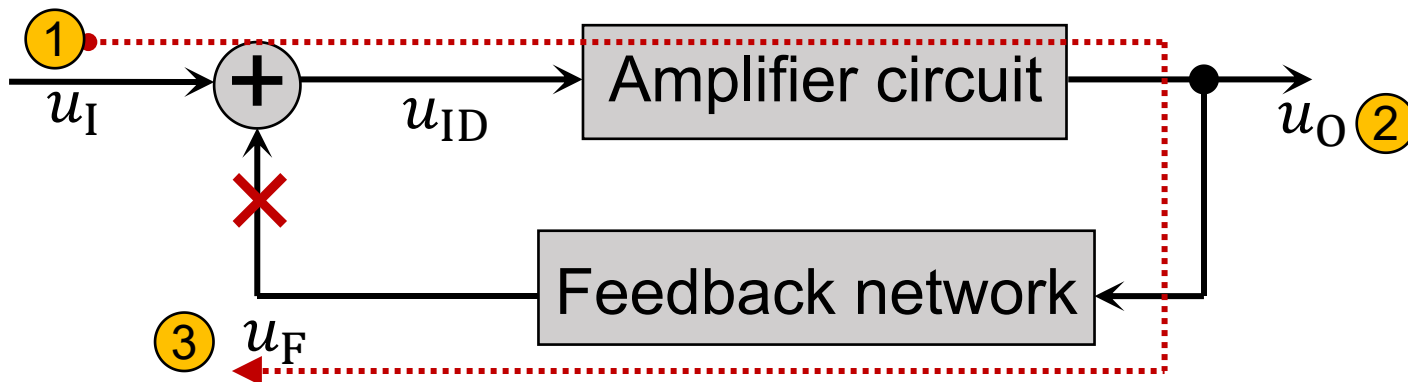
- The net input signal $<$ input signal: **negative feedback**

Improve the performance of amplifier circuit

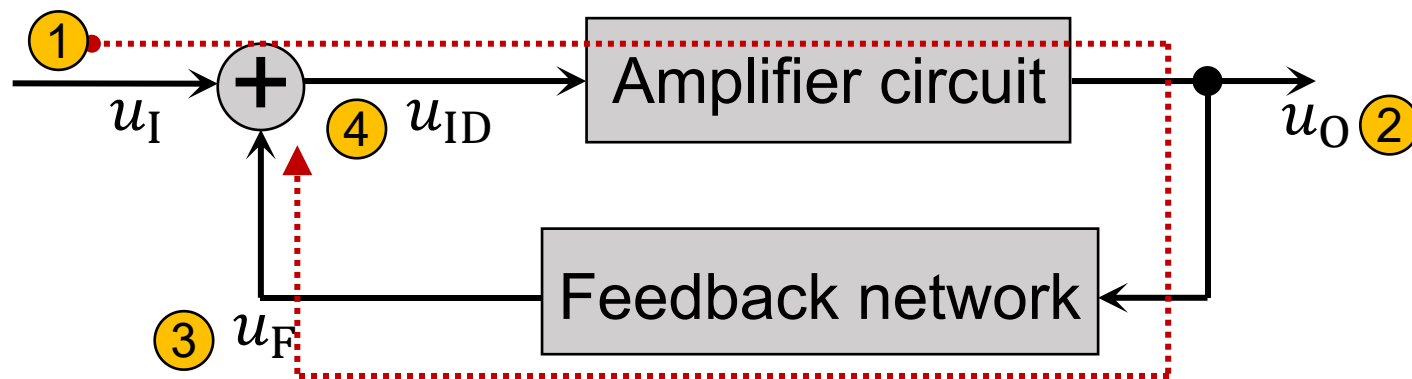
Signal operation 信号运算

How to judge the positive and negative feedback

(1) Apply an instantaneous positive value to u_I 给出入信号加上一个瞬时的正信号



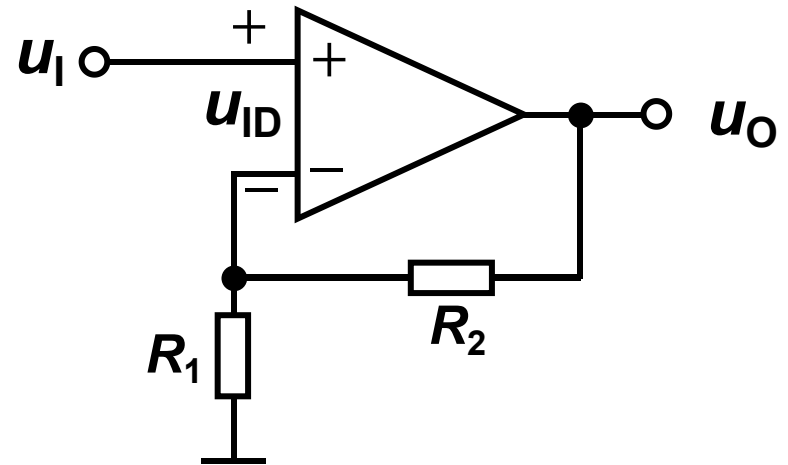
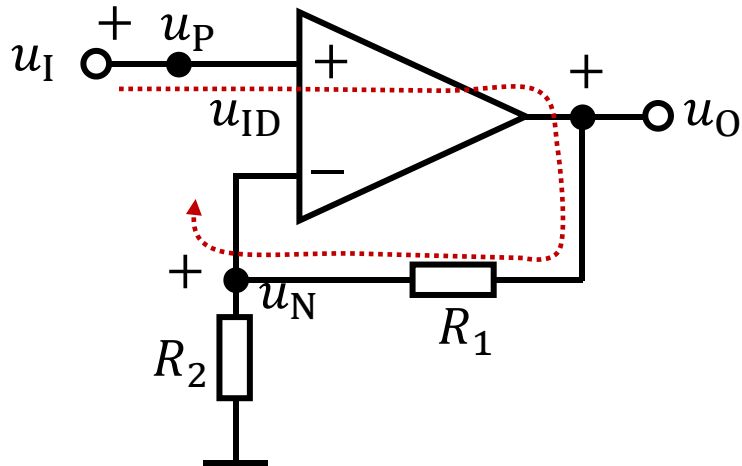
(2) Follow the red guideline (open the feedback branch), firstly determine the polarity of u_O , then determined the polarity of u_F 断开反馈支路，依次判断 u_O 和 u_F 的极性



(3) Close the feedback branch. $u_{ID} < u_I$: Negative feedback; $u_{ID} > u_I$: Positive feedback;

Example 1: Positive or negative feedback?

(1) Assume u_I is positive to ground

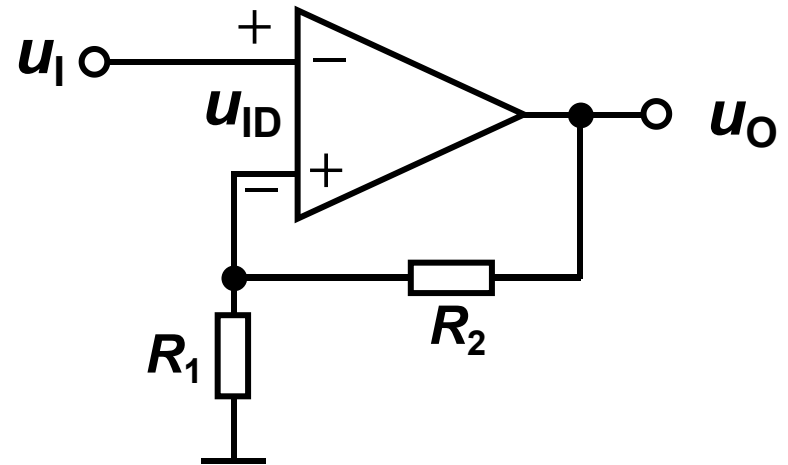
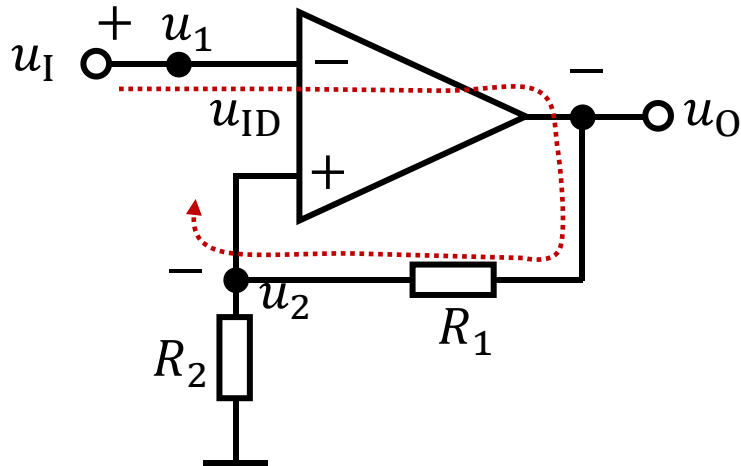


(2) $u_O: + \Rightarrow u_N: +$,

(3) $u_{ID} = u_P - u_N < u_I \Rightarrow$ Negative feedback

Example 2: Positive or negative feedback?

(1) Assume u_I is positive to ground



(3) $u_O: - \rightarrow u_2: -$,

(4) $u_{ID} = u_1 - u_2 > u_I \rightarrow$ Positive feedback

Example 3: Positive or negative feedback for AC circuit?

(1) Draw the AC circuit

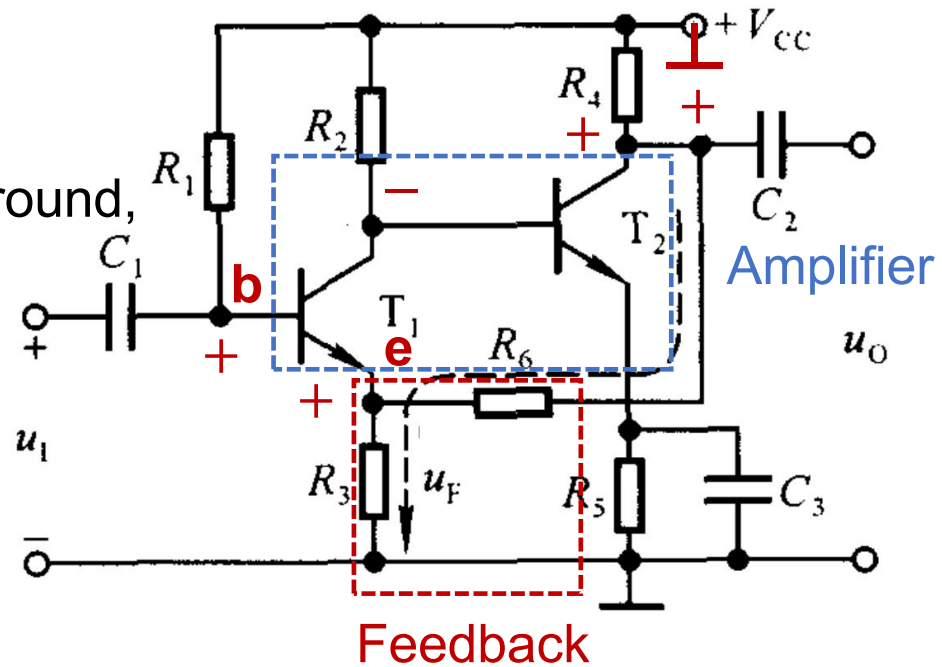
(2) Find amplifier circuit and feedback circuit

(3) Assume u_I is positive to ground, open the feedback circuit

(4) $u_b: + \rightarrow u_e: +$,

(5) $u_{be} = u_b - u_e < u_I$

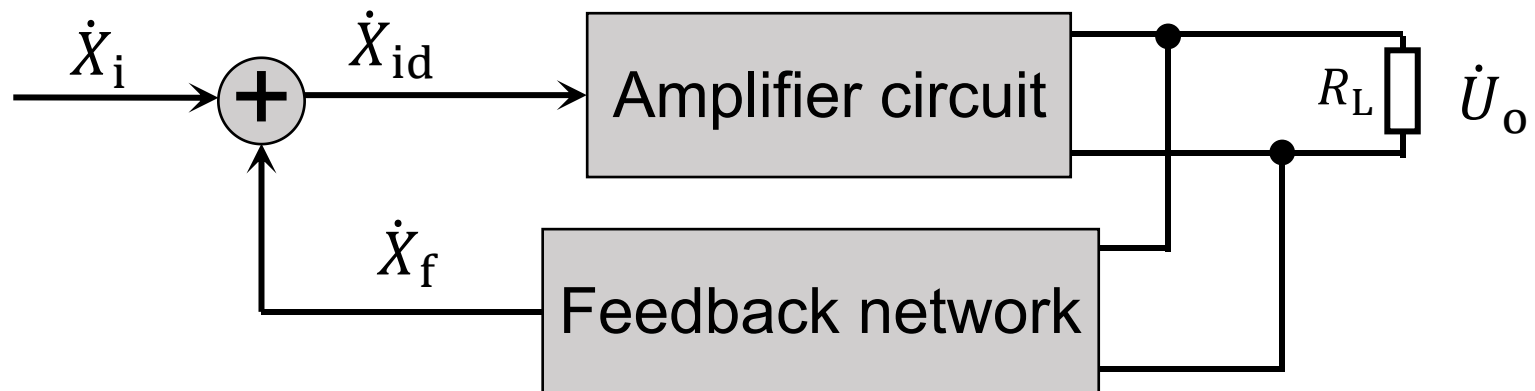
$\rightarrow$ Negative feedback



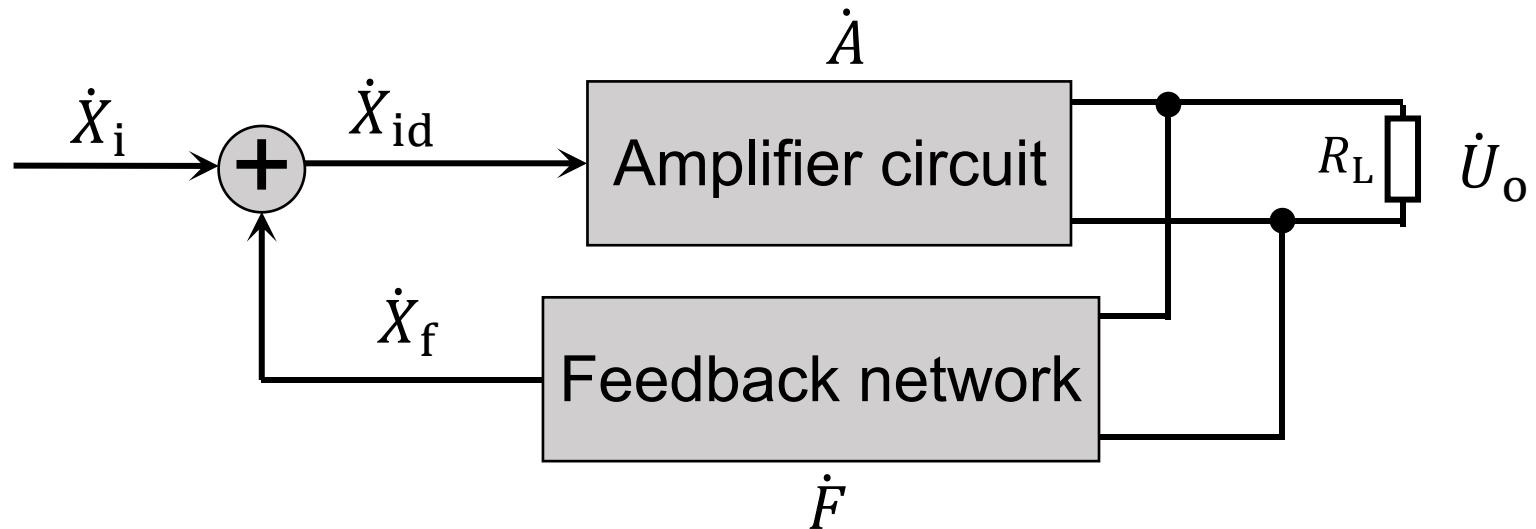
6.2 AC negative feedback amplifier circuits

Output terminal { Voltage feedback 电压反馈
输出端 { Current feedback 电流反馈

Voltage feedback 电压反馈

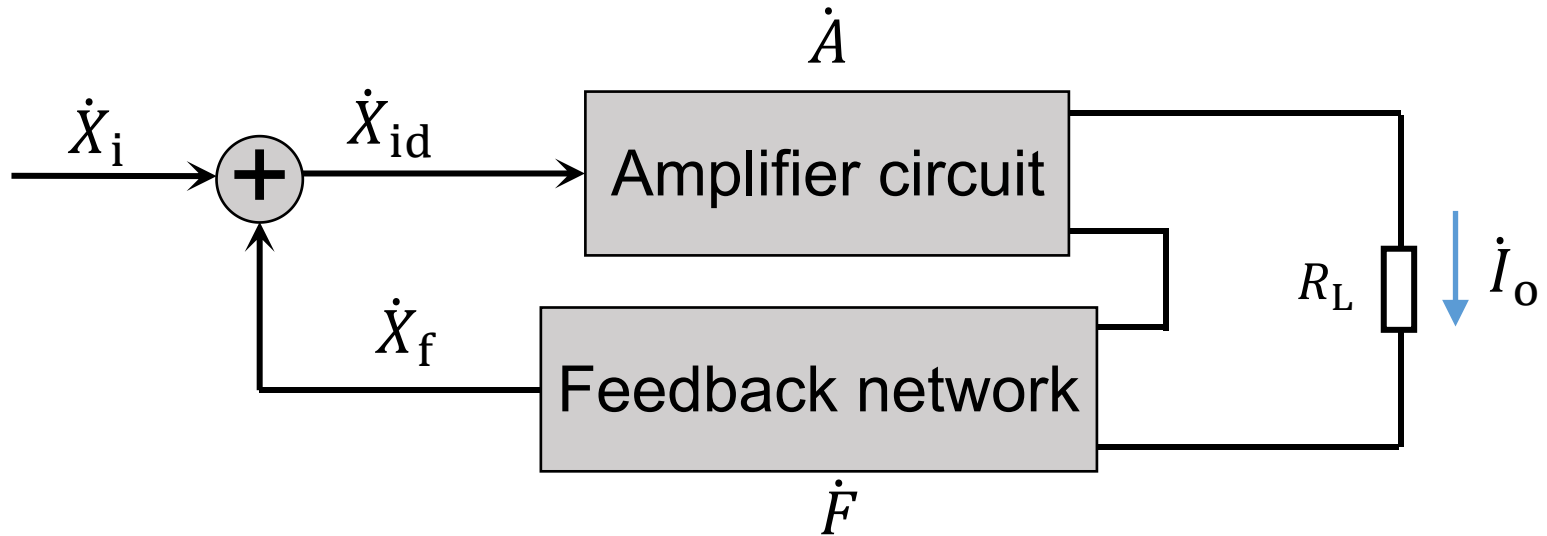


Voltage feedback 电压反馈



- The feedback network is in parallel connection with R_L
 - The feedback signal is proportional to the output voltage
- $$\dot{X}_f = \dot{F} \dot{U}_o$$

Current feedback



- The feedback network is in series connection with R_L
- The feedback signal is proportional to the output current

$$\dot{X}_f = \dot{F} i_o$$

How to judge the voltage and current feedback?

First approach (the most convenient approach):

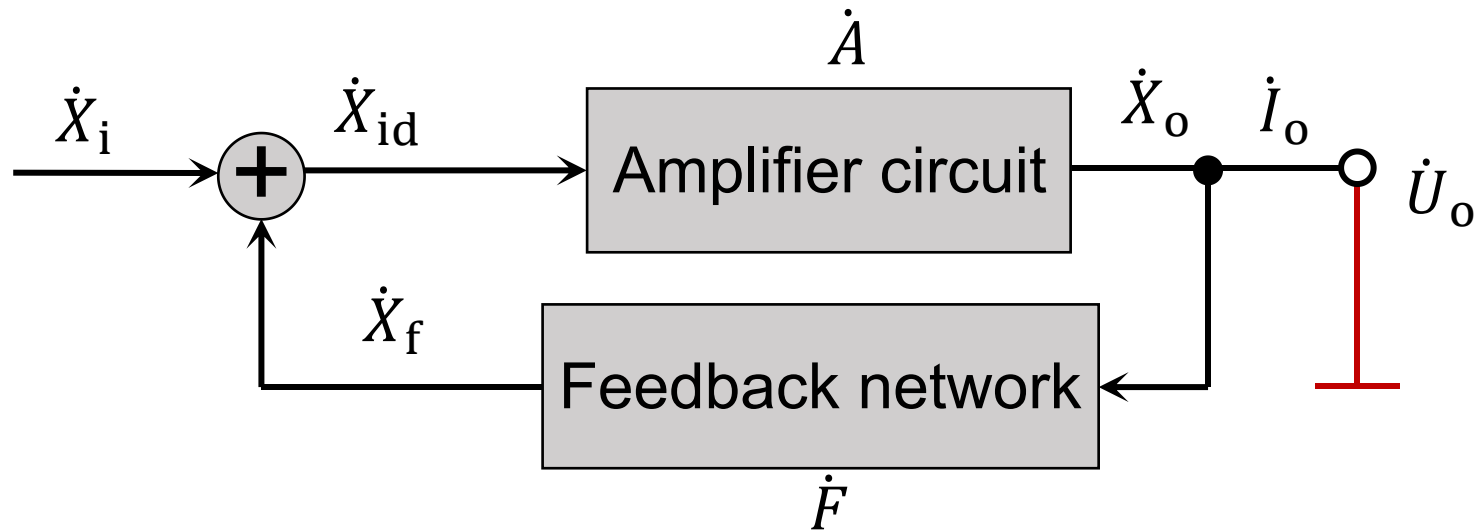
❑ If feedback network in **parallel connection** with load resistor, it is voltage feedback.

Input signal for feedback network is voltage.

❑ If feedback network in **series connection** with load resistor, it is current feedback.

Input signal for feedback network is current.

Second approach (be careful with this approach):



Short the load resistor (connect output terminal to ground):

$$\dot{U}_o = 0 \quad \left\{ \begin{array}{l} \text{If } \dot{X}_f = \dot{F}\dot{U}_o = 0, \text{ it is voltage feedback} \\ \text{If } \dot{X}_f = \dot{F}\dot{U}_o \neq 0, \text{ it is current feedback} \end{array} \right.$$

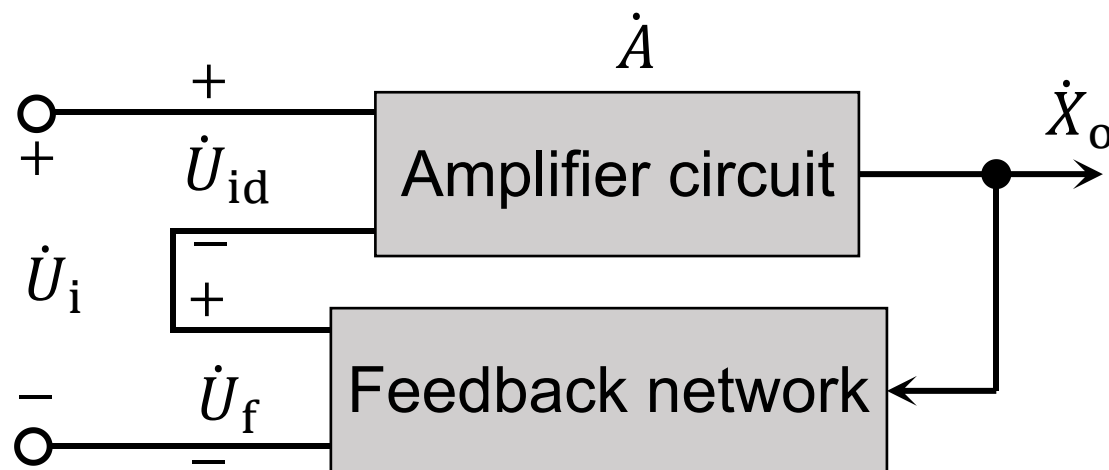
In some special cases, part of input signal $\dot{X}_i$ will flow through feedback branch and make the total signal value on feedback branch ($\dot{X}_f + \dot{X}_{ipart}$) is not zero even it is voltage feedback when $\dot{U}_o = 0$.

Input terminal

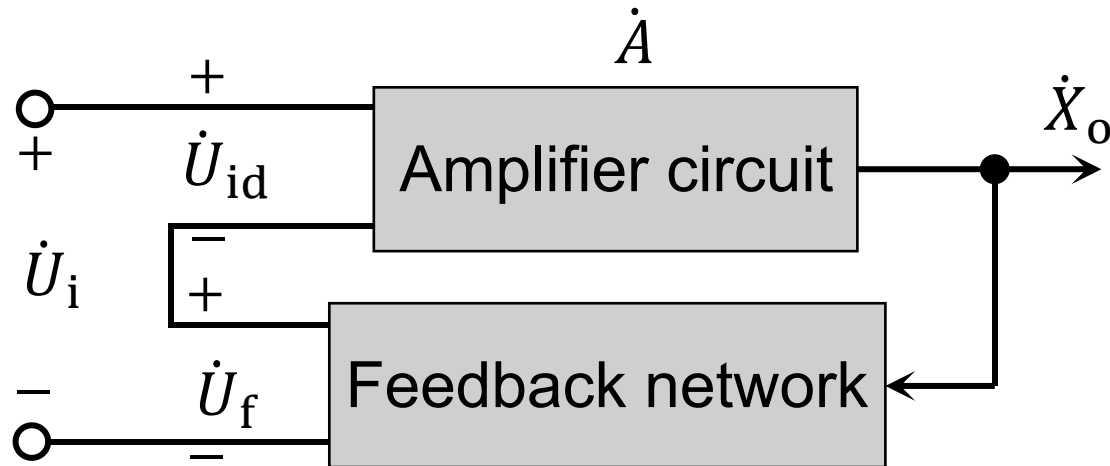
输入端

{ Series feedback 串联反馈
Parallel feedback 并联反馈

Series feedback

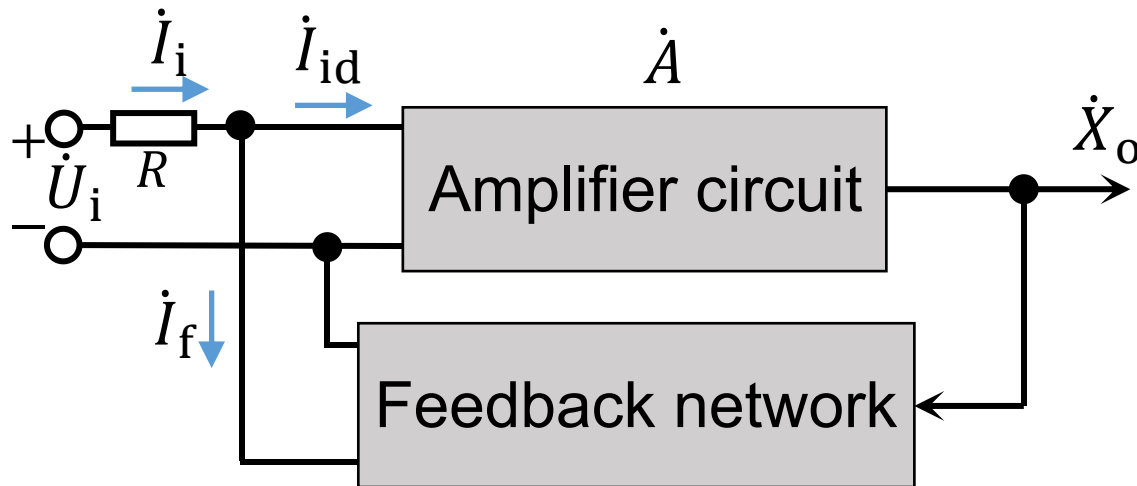


Series feedback



- The feedback network is in series connection with the net input circuit $\dot{U}_i = \dot{U}_{id} + \dot{U}_f$
- The feedback signal is voltage signal

Parallel feedback

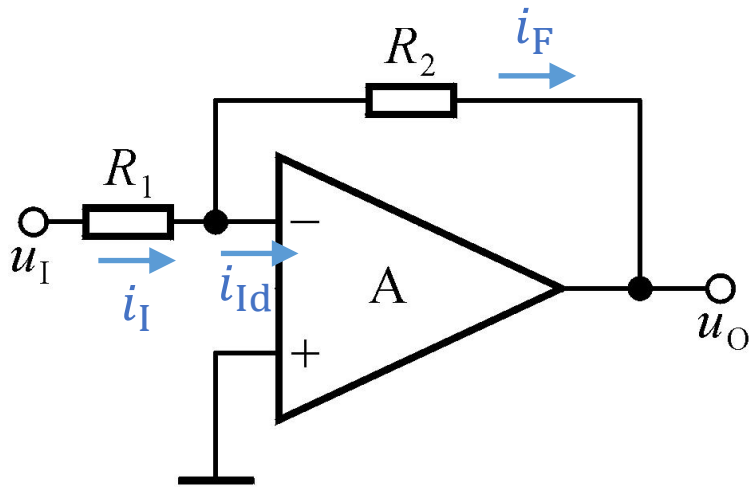


The feedback network is in parallel connection with the input circuit

$$\dot{I}_i = \dot{I}_{id} + \dot{I}_f$$

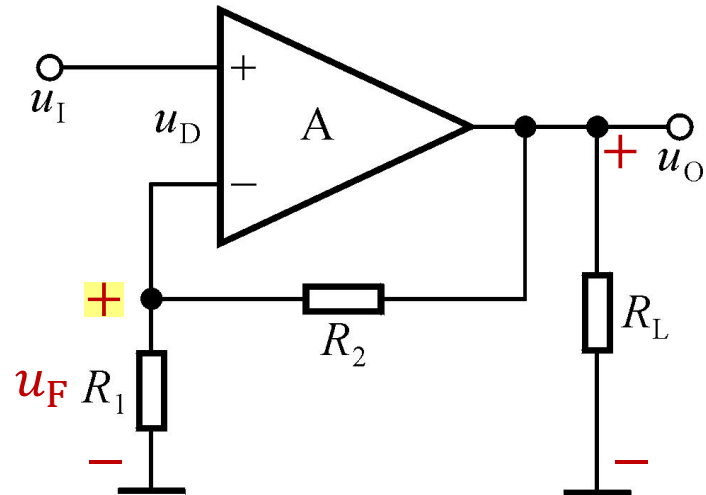
The feedback signal is current signal

How to judge the series and parallel feedback?



$$i_I = i_{Id} + i_F$$

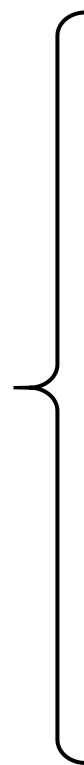
Parallel feedback



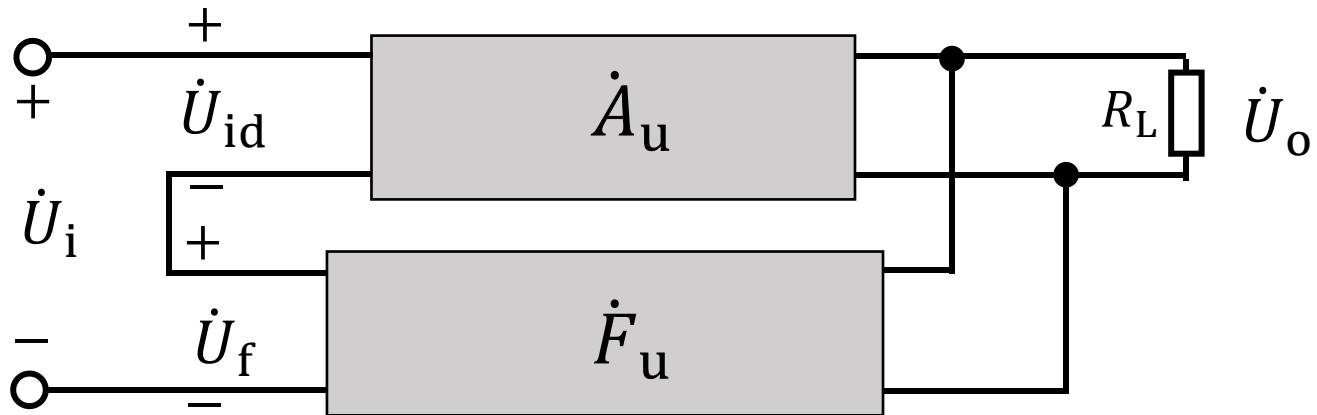
$$u_I = u_D + u_F$$

Series feedback

In total, there are 4 type AC negative feedback circuits

- 
- Voltage series** feedback 电压串联反馈
 - Voltage parallel feedback 电压并联反馈
 - Current series feedback 电流串联反馈
 - Current parallel** feedback 电流并联反馈

Voltage series feedback



$$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_{id}}$$

Open loop gain

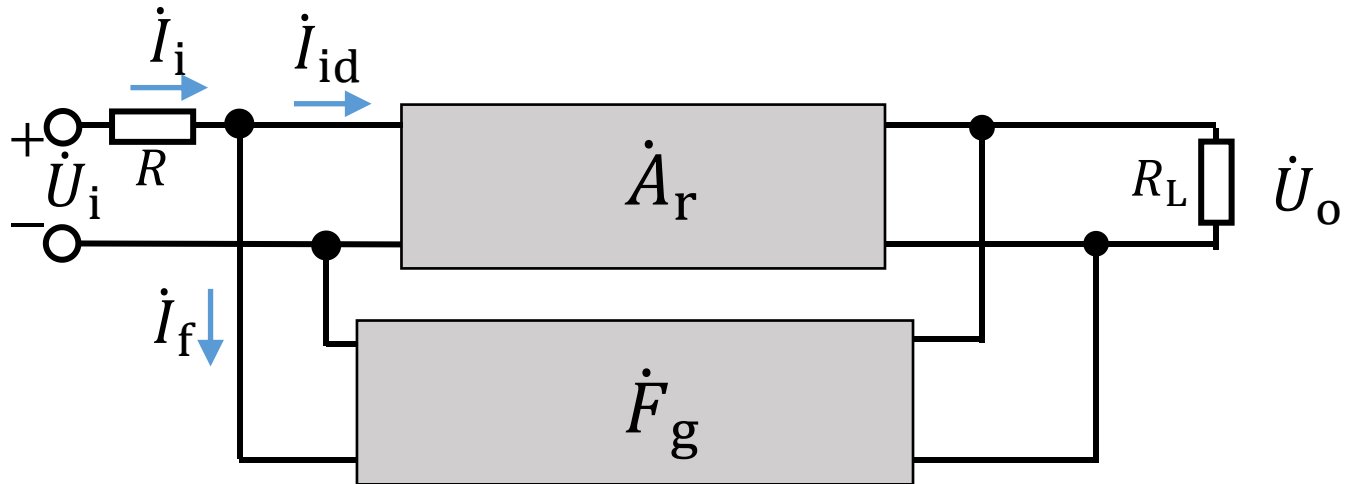
$$\dot{F}_u = \frac{\dot{U}_f}{\dot{U}_o}$$

Feedback
factor

$$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_i}$$

Close loop gain

Voltage parallel feedback



$$\dot{A}_r = \frac{\dot{U}_o}{\dot{I}_{id}}$$

Open loop gain

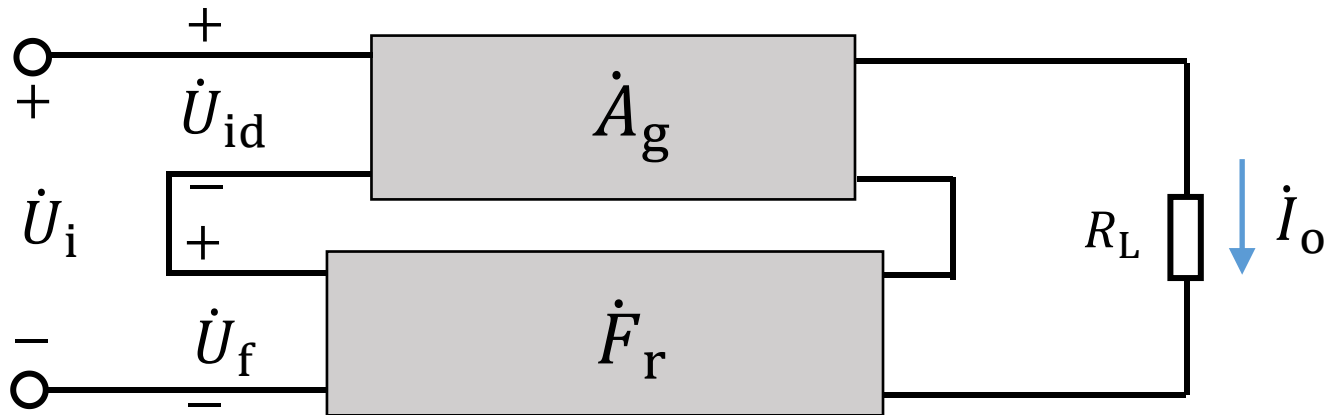
$$\dot{F}_g = \frac{\dot{I}_f}{\dot{U}_o}$$

Feedback
factor

$$\dot{A}_{rf} = \frac{\dot{U}_o}{\dot{I}_i}$$

Close loop gain

Current series feedback



$$\dot{A}_g = \frac{\dot{I}_o}{\dot{U}_{id}}$$

Open loop gain

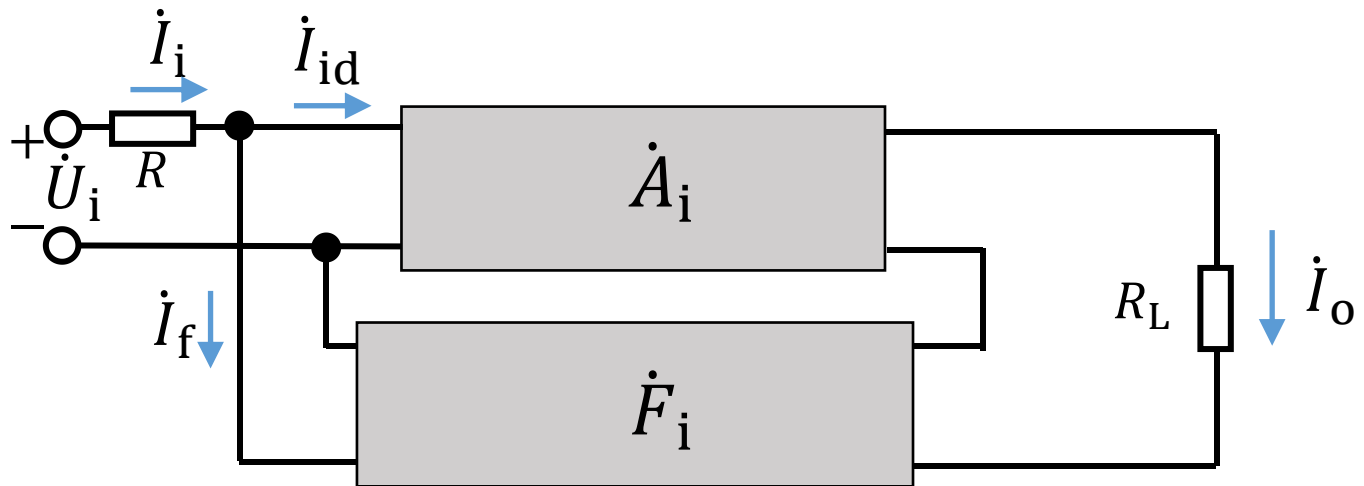
$$\dot{F}_r = \frac{\dot{U}_f}{\dot{I}_o}$$

Feedback
factor

$$\dot{A}_{gf} = \frac{\dot{I}_o}{\dot{U}_i}$$

Close loop gain

Current parallel feedback



$$\dot{A}_i = \frac{\dot{I}_o}{\dot{I}_{id}}$$

Open loop gain

$$\dot{F}_i = \frac{\dot{I}_f}{\dot{I}_o}$$

Feedback factor

$$\dot{A}_{if} = \frac{\dot{I}_o}{\dot{I}_i}$$

Close loop gain

Summary of 4 type AC negative feedback circuits

	$\dot{A}$	$\dot{F}$	$\dot{A}_f$	Functionality
Voltage series	$\dot{A}_u = \frac{\dot{U}_o}{\dot{U}_{id}}$	$\dot{F}_u = \frac{\dot{U}_f}{\dot{U}_o}$	$\dot{A}_{uf} = \frac{\dot{U}_o}{\dot{U}_i}$	Voltage amplification
Voltage parallel	$\dot{A}_r = \frac{\dot{U}_o}{\dot{I}_{id}}$	$\dot{F}_g = \frac{\dot{I}_f}{\dot{U}_o}$	$\dot{A}_{rf} = \frac{\dot{U}_o}{\dot{I}_i}$	Convert current to voltage
Current series	$\dot{A}_g = \frac{\dot{I}_o}{\dot{U}_{id}}$	$\dot{F}_r = \frac{\dot{U}_f}{\dot{I}_o}$	$\dot{A}_{gf} = \frac{\dot{I}_o}{\dot{U}_i}$	Convert voltage to current
Current parallel	$\dot{A}_i = \frac{\dot{I}_o}{\dot{I}_{id}}$	$\dot{F}_i = \frac{\dot{I}_f}{\dot{I}_o}$	$\dot{A}_{if} = \frac{\dot{I}_o}{\dot{I}_i}$	Current amplification

Example 1:

(1) Negative or positive feedback for AC circuit?

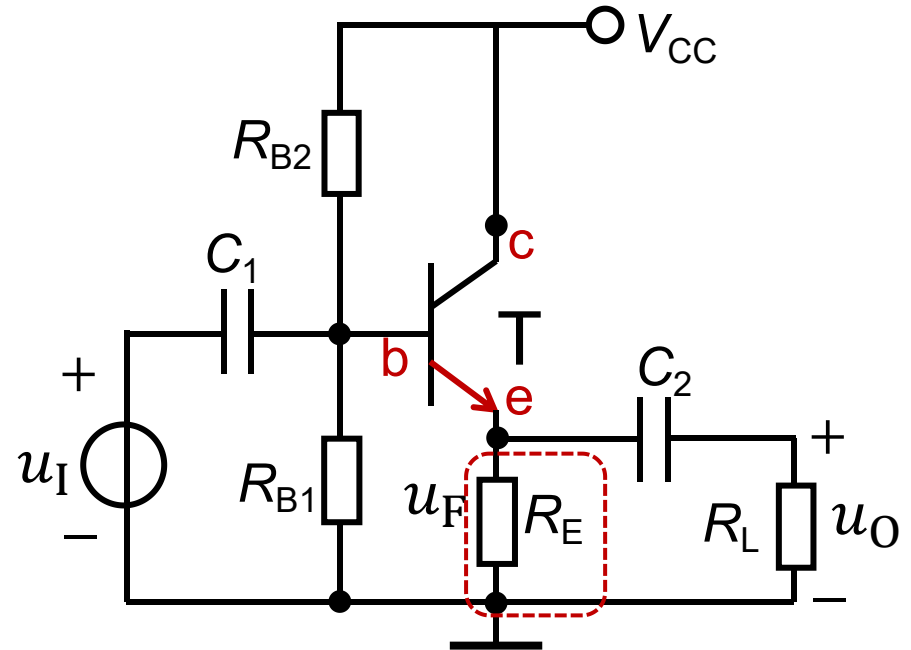
Assume: $u_I > 0$

$$u_b > 0$$

$$u_F = u_e > 0$$

$$u_{Id} = u_{be} = u_I - u_F < u_I$$

Negative feedback !



Example 1:

(2) Judge the type of the negative feedback circuit

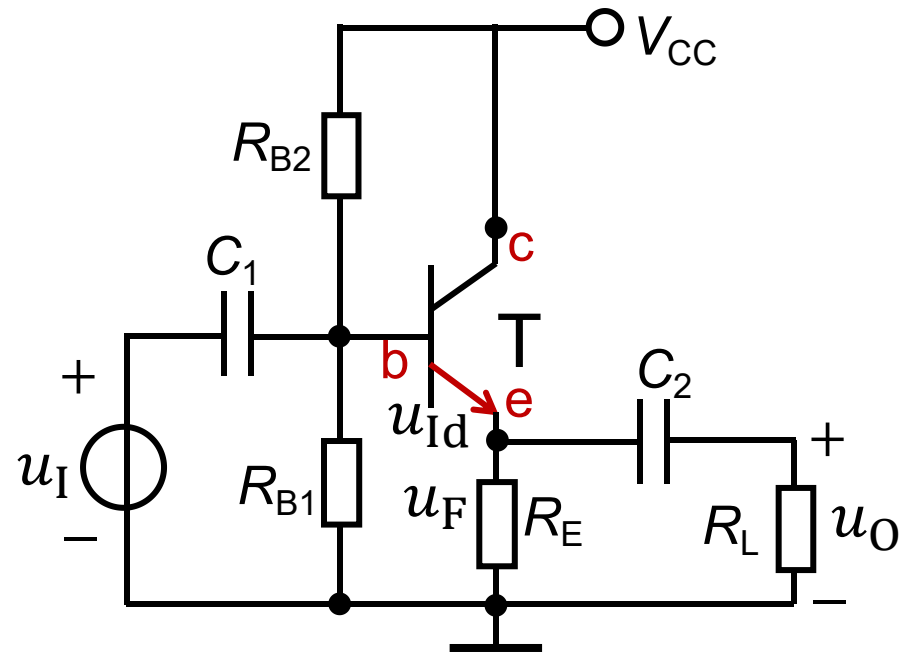
When $u_O = 0$, $u_F = 0$

Voltage feedback!

$$\therefore u_I = u_{be} + u_F$$

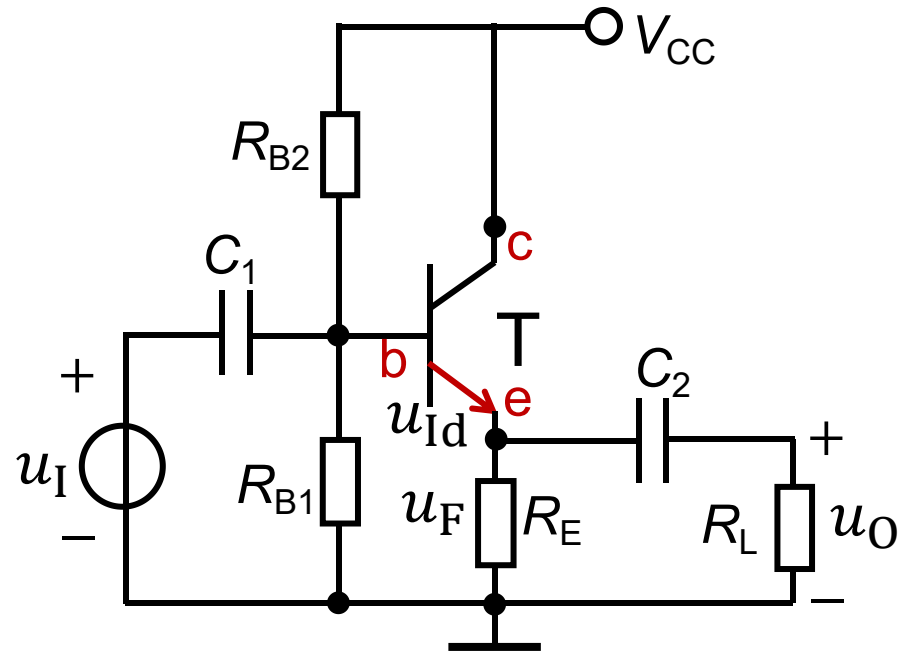
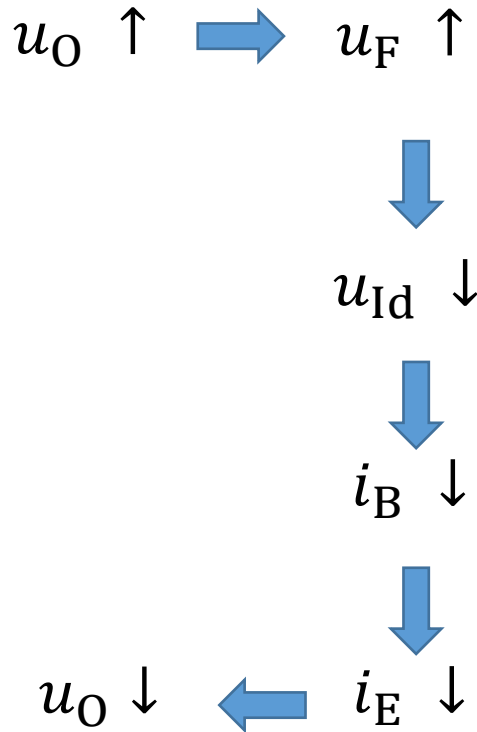
Series feedback!

Voltage series feedback!



Example 1:

(3) The working principle of stabilizing the output voltage



Example 2:

(1) Negative or positive feedback for AC circuit?

Assume: $u_I > 0$

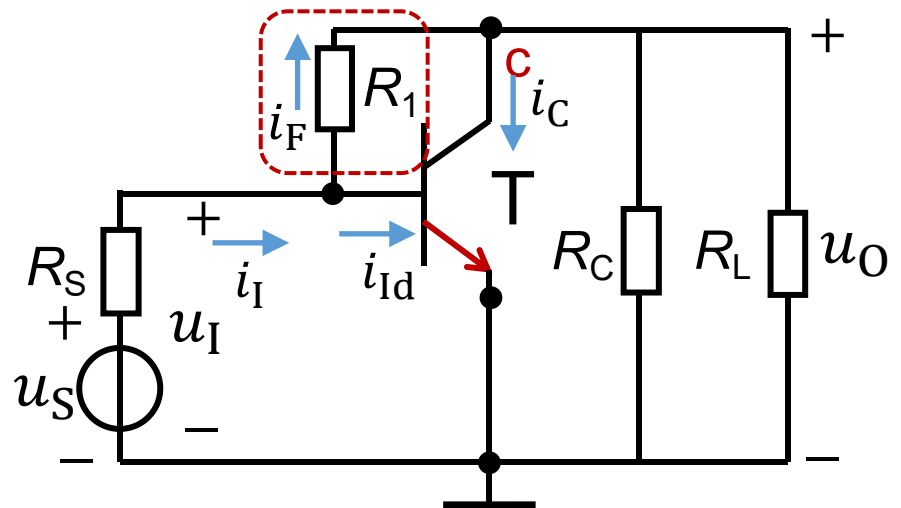
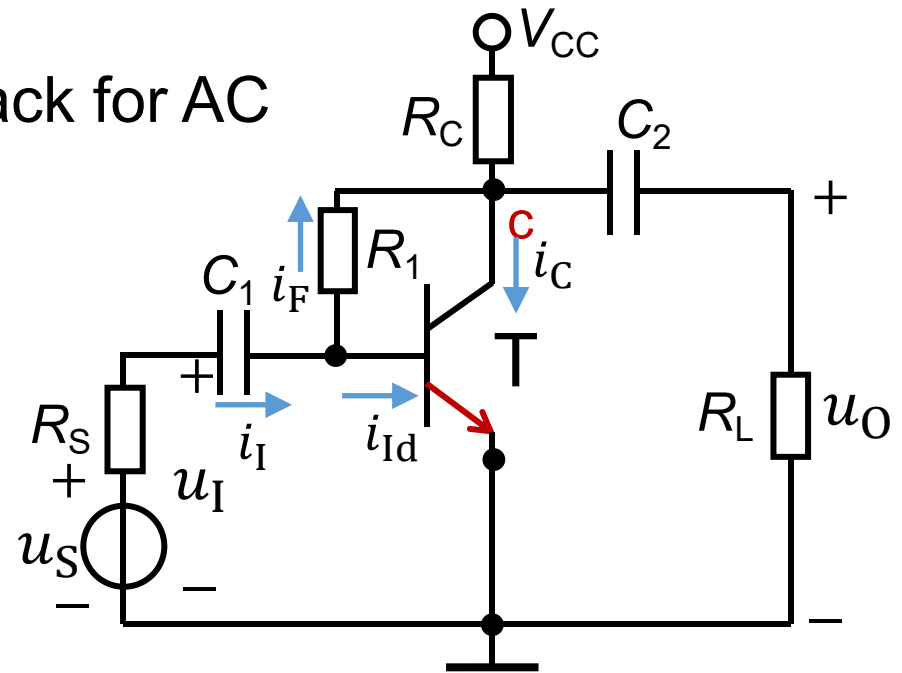
$$i_I > 0, \quad i_{Id} > 0, \quad i_C > 0$$

$$u_C < 0$$

$$i_F > 0$$

$$i_{Id} < i_I$$

Negative feedback !

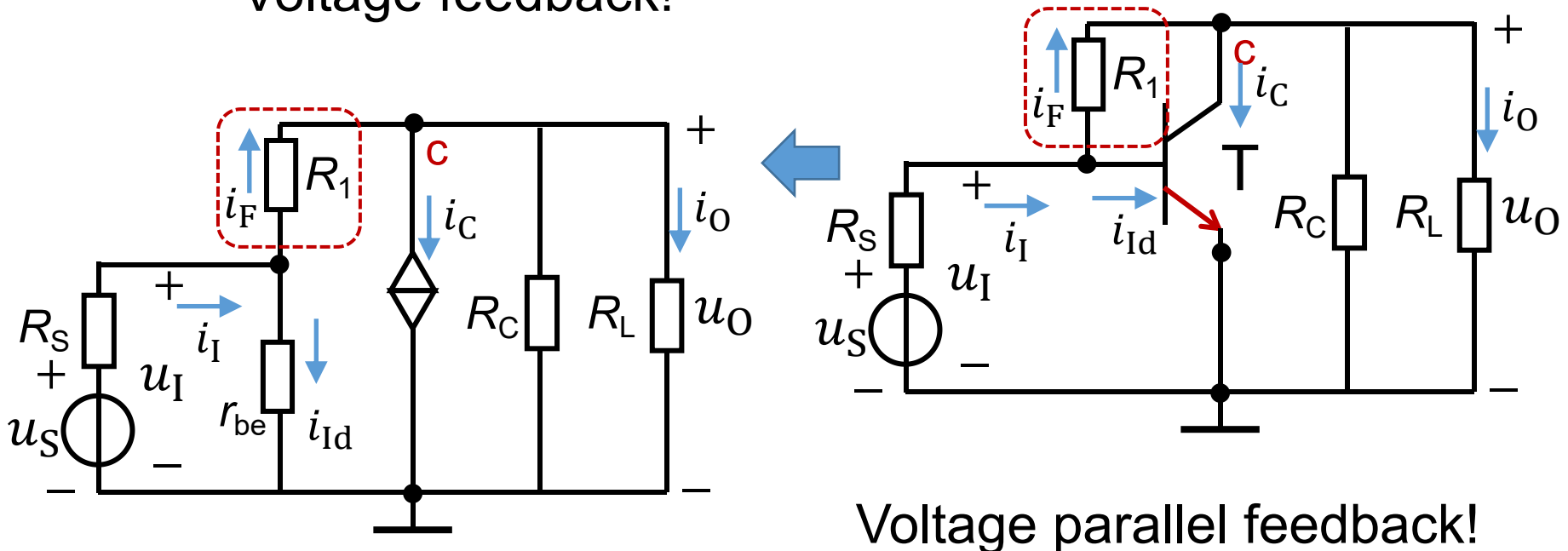


Example 2:

(2) Judge the type of the negative feedback circuit

R_1 is in parallel connection with R_L

Voltage feedback!



$R_1 + (R_S || r_{be})$ is in parallel connection with output terminal. Hence, part of u_O will be transmitted back to R_1 . It is voltage feedback.

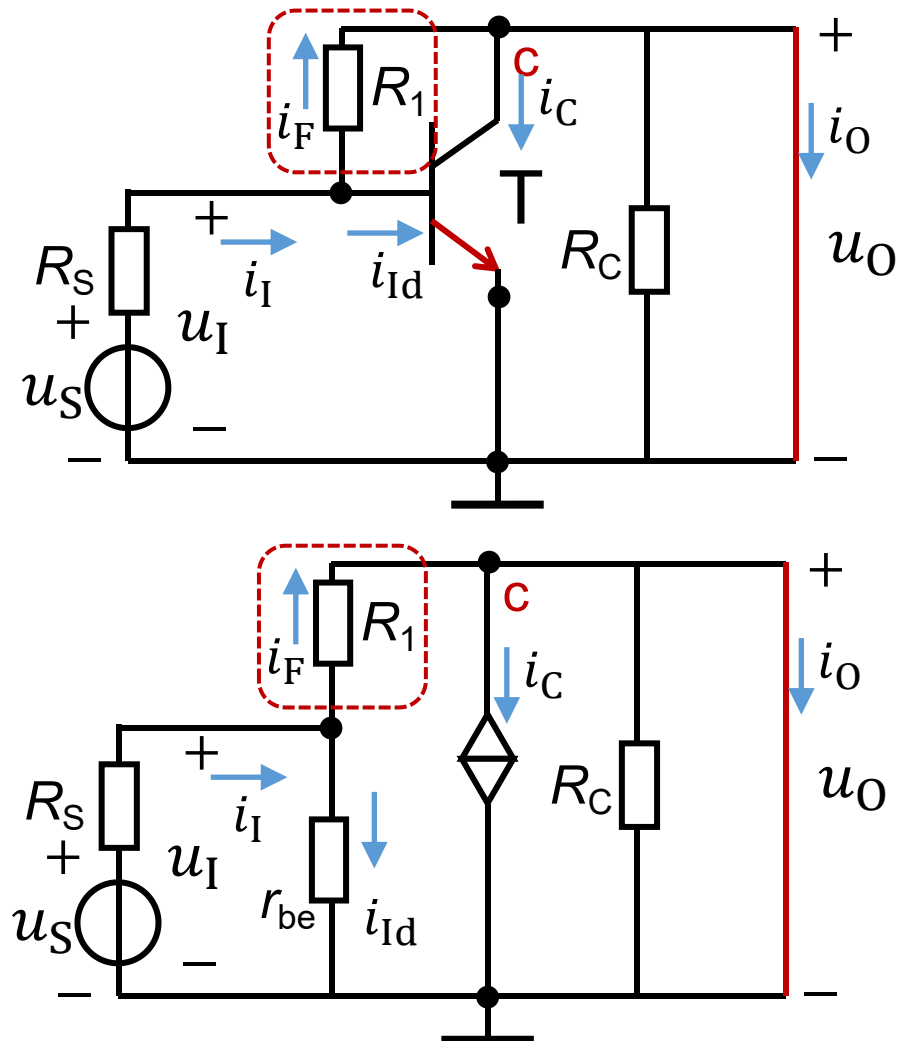
Example 2:

(2) Judge the type of the negative feedback circuit

When $R_L = 0$, $u_O = 0$: $i_F \neq 0$

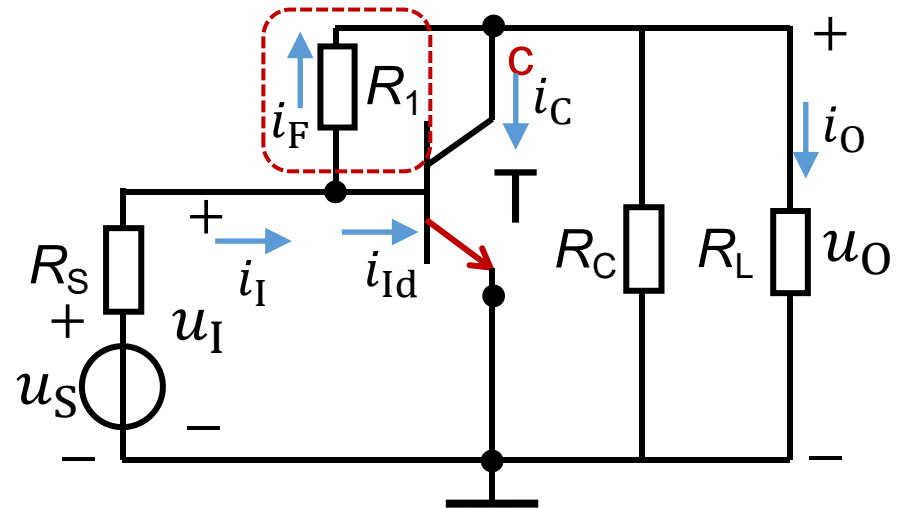
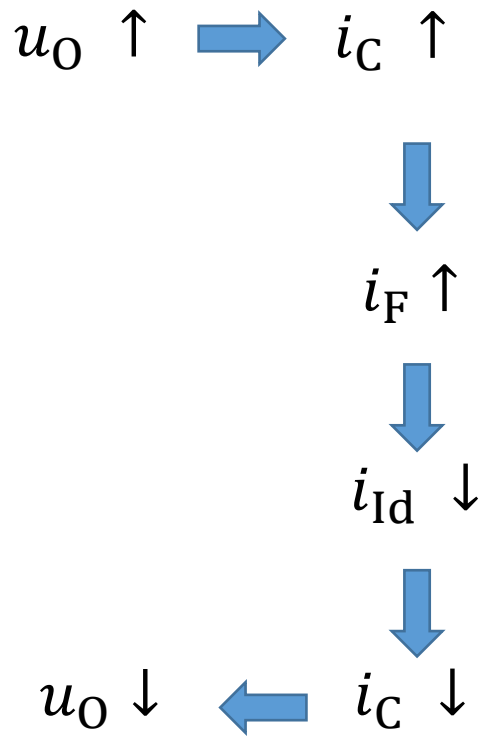
i_F { From $\dot{X}_f = \dot{F}\dot{U}_o = 0$
From $i_I \neq 0$

Voltage feedback!



Example 2:

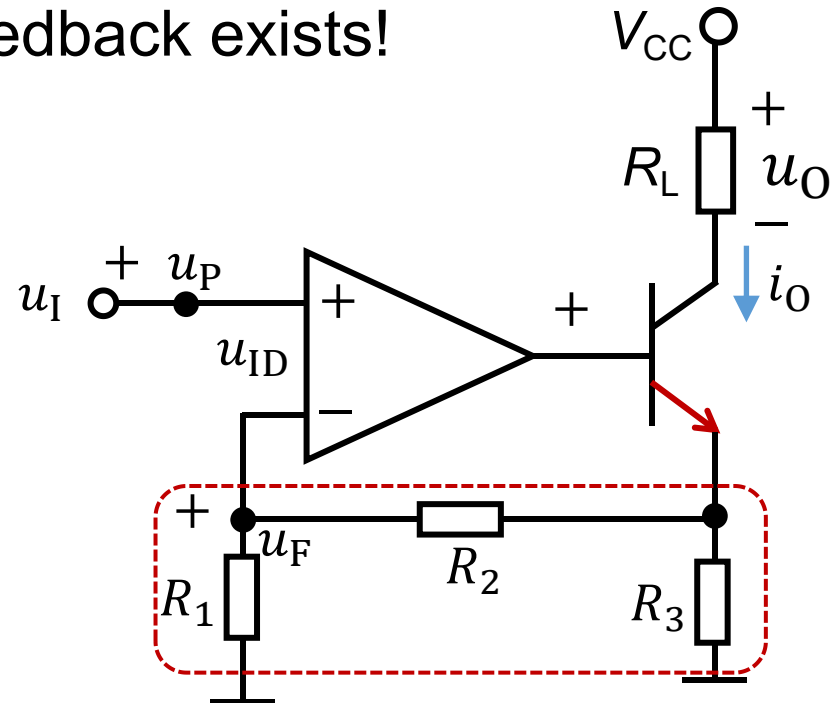
(3) The working principle of stabilizing the output voltage



Example 3:

(1) DC or AC feedback?

Both DC feedback and AC feedback exists!



Example 3:

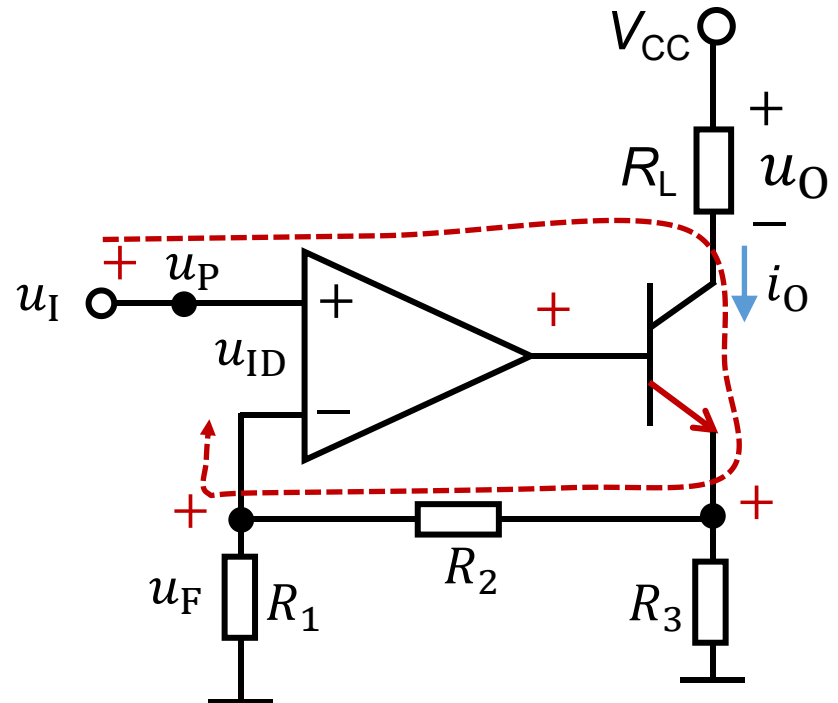
(2) Negative or positive feedback for AC circuit?

Assume: $u_I > 0$

$$u_F > 0$$

$$u_{ID} < u_I$$

Negative feedback!



Example 3:

(3) The type of AC negative feedback

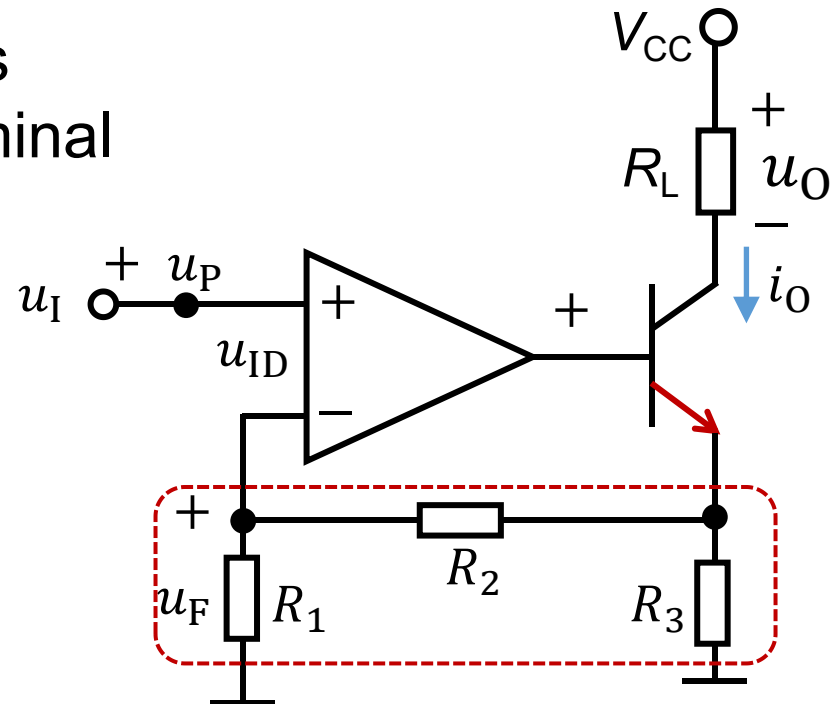
Feedback network in series connection with output terminal

Current feedback!

$$\because u_I = u_{ID} + u_F$$

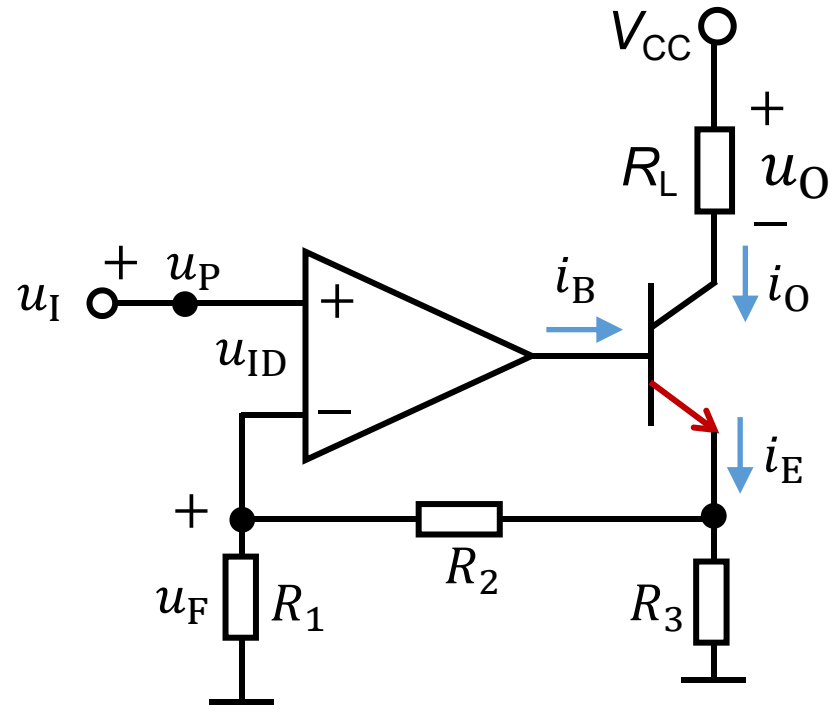
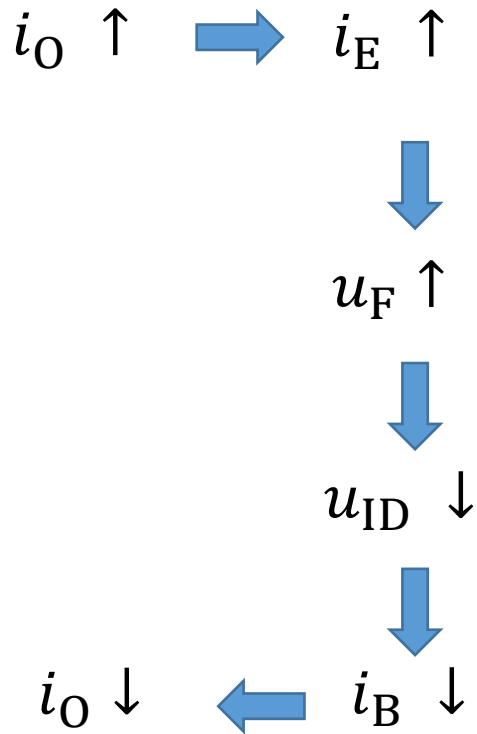
Series feedback!

Current series feedback!



Example 3:

(4) The working principle of stabilizing the output current



Homework 6: Refer to blackboard

Deadline: 2025-05-14 11:59pm